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PAYLOAD DELIVERY AND DISPERSAL PROJECTILE

Abstract

A projectile is launched from a projectile launcher to disperse a projectile payload proximate to a target. The projectile is comprised of a projectile housing, a primer assembly, and a primer activator. The projectile payload and the primer assembly may be disposed within a housing channel of the projectile housing. The primer assembly is comprised of a primer charge configured to deploy the projectile payload from the projectile housing and a primer activator configured to activate the primer charge. The primer activator may activate the primer charge in response to the first end of the projectile housing contacting a target. Further, the primer charge may cause the projectile payload to be dispersed from the projectile.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application 63/471,437 filed Jun. 6, 2023, which is incorporated by reference herein in its entirety.

FIELD OF THE INVENTION
[0002] Embodiments of the present disclosure relate to a projectile launcher.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] FIG. **1** is a perspective view of a projectile launcher, in accordance with various embodiments;
[0004] FIG. **2** is a schematic view of a projectile launcher, in accordance with various embodiments;
[0005] FIG. **3**A is a front perspective view of a projectile launcher magazine, in accordance with various embodiments;
[0006] FIG. **3**B is a rear perspective view of a projectile launcher magazine, in accordance with various embodiments;
[0007] FIG. **4** is a cross section view of a projectile for a projectile launcher, in accordance with various embodiments;
[0008] FIG. **5**A is a cross section view of a projectile for a projectile launcher in a first state, in accordance with various embodiments;
[0009] FIG. **5**B is a cross section view of a projectile for a projectile launcher in a second state, in accordance with various embodiments;
[0010] FIG. **5**C is a cross section view of a projectile for a projectile launcher in a third state, in accordance with various embodiments;
[0011] FIG. **6** is a cross section view of a projectile for a projectile launcher, in accordance with various embodiments;
[0012] FIG. **7**A is a cross section view of a first projectile that deploys a projectile payload, in accordance with various embodiments;
[0013] FIG. **7**B is a cross section view of a second projectile that deploys a projectile payload, in accordance with various embodiments; and
[0014] FIG. **7**C is a cross section view of a third projectile that deploys a projectile payload, in accordance with various embodiments.
DETAILED DESCRIPTION
[0015] The detailed description of exemplary embodiments herein makes reference to the accompanying drawings, which show exemplary embodiments by way of illustration. While these embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosures, it should be understood that other embodiments may be realized and that logical changes and adaptations in design and construction may be made in accordance with this disclosure and the teachings herein. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation.
[0016] The scope of the disclosure is defined by the appended claims and their legal equivalents rather than by merely the examples described. For example, the steps recited in any of the method or process descriptions may be executed in any order and are not necessarily limited to the order presented. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any

reference to attached, fixed, coupled, connected, or the like may include permanent, removable, temporary, partial, full, and/or any other possible attachment option. Surface shading lines may be used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

[0017] Systems, methods, and apparatuses may be used to interfere with voluntary locomotion (e.g., walking, running, moving, etc.) of a target. For example, a projectile launcher may be configured as a conducted electrical weapon ("CEW") may be used to deliver a current (e.g., stimulus signal, pulses of current, pulses of charge, etc.) through tissue of a human or animal target. Although generally referred to as a CEW, as described herein, the term CEW may refer to a conducted electrical weapon, a conducted energy weapon, and/or any other similar device or apparatus configured to provide a stimulus signal through one or more deployed projectiles (e.g., electrodes).

[0018] Systems, methods, and apparatuses may be used to interfere with, prevent, and/or otherwise disincentivize escalation and/or further interaction during an event. For example, a projectile launcher may be utilized to deploy a deterrent and/or other substance to a human or animal target that disincentivizes further interaction. Additionally, the projectile launcher may be utilized to deploy an identifying substance and/or device to a human or animal target that enables the human or animal target to be identified at a different location and/or a later time removed from the event. The deterrent, identifying substance, and/or other substances may be referred to as a projectile payload.

[0019] A projectile payload may comprise a substance selected to induce a desired response and/or enable further interaction at a safe location and appropriate time. The substance may be stored and/or provided to the targeted human and/or animal as a gas, liquid, powered solid, fluid, and/or an otherwise dispersible state. Where the projectile payload comprises a deterrent, the projectile launcher may launch a projectile to disperse the projectile payload in proximity to a human or animal target. During dispersal, the deterrent may contact and/or otherwise interact with the human or animal target to cause pain, discomfort, confusion, disorientation, and/or otherwise deter the human or animal target from continuing a course of action. Where the projectile payload comprises other substances, the projectile launcher may launch the projectile to disperse and/or otherwise deploy the projectile payload in proximity to and/or in contact with the human or animal target.

[0020] A magazine may be a housing and/or structural component that receives one or more projectiles and/or cartridges. The magazine may be configured to receive and/or secure the one or more projectiles (and/or cartridges) within one or more firing tubes. Additionally, the magazine may be configured to fit within and/or couple with a handle of a projectile launcher. In some embodiments, each projectile may be directly received in a magazine. For example, the magazine may receive a respective projectile within the one or more firing tubes prior to deployment. After a projectile is deployed, another projectile may be inserted in the magazine to permit launch of another projectile. Projectiles may be inserted within the magazine while the magazine is associated with the handle or when the magazine is unassociated with the handle. Alternately or additionally, the magazine may receive one or more cartridges that each include one or more projectiles. Similar to the respective projectile above, the magazine may receive a respective cartridge within the one or more firing tubes prior to deployment.

[0021] In various embodiments, a magazine may include two or more projectiles (e.g., a cartridge containing two or more projectiles, two or more projectiles are launched directly from two or more firing tubes of the magazine, etc.) that are launched at the same time. In various embodiments, a magazine may include two or more projectiles that may each be launched individually at separate times. In various embodiments, a magazine may include a single projectile configured to be launched from the magazine. Launching the projectiles may be referred to as activating (e.g., firing) a magazine and/or a bay of the handle. After use (e.g., activation, firing), a magazine may be removed from the bay and replaced with an unused (e.g., not fired, not activated) magazine to permit launch of additional projectiles.

[0022] A cartridge may be configured as an external housing comprised of a single projectile. For example, a CEW cartridge may comprise a single electrode. Alternatively, a payload cartridge may comprise a single projectile having a single payload. The single projectile may be individually deployed from the cartridge and the magazine in which the cartridge is received. Alternately, a cartridge may comprise two or more projectiles that are launched at the same time. Launching or deploying a projectile may be referred to as activating (e.g., firing) a cartridge. After use (e.g., activation, firing), at least a portion of a cartridge may remain in the magazine. After the use, the portion of the cartridge may be removed from the magazine and replaced with an unused (e.g., not fired, not activated) cartridge to permit launch of an additional projectile or projectiles.

[0023] In various embodiments, a projectile launcher may include a handle and one or more magazines (e.g., deployment units, etc.). The handle may include one or more bays for receiving the magazine(s). Each magazine may be removably positioned in (e.g., inserted into, coupled to, etc.) a bay. Each magazine may releasably electrically, electronically, and/or mechanically couple to a bay. A deployment of the projectile launcher may launch one or more projectiles from the magazine toward a target to remotely deliver a projectile payload.

[0024] In various embodiments, a projectile may include a handle and one or more bays for receiving one or more projectiles (e.g., electrodes, pepper projectiles, marker projectiles, etc.). Each projectile may be removably positioned in (e.g., inserted into, coupled to, etc.) a bay. Each projectile may releasably electrically, electronically, and/or mechanically couple to a bay. A deployment of the projectile launcher may launch one or more projectiles from one or more bays of the handle and toward a target to remotely deliver the stimulus signal and/or the projectile payload.

[0025] In various embodiments, and with reference to FIG. **1**, a projectile launcher **100** is disclosed. Projectile launcher **100** may be similar to, or have similar aspects and/or components with, any projectile launcher discussed herein. Projectile launcher **100** may comprise a housing **102** and a magazine **104**. The housing **102** of projectile launcher **100** may further comprise a magazine receiver **106**, a trigger **108**, a control interface **110**, a handle end **112**, and a deployment end **114**. It should be understood by one skilled in the art that FIGS. **3**A and **3**B are schematic representations of projectile launcher **100**. It should be further noted that any of the one or more components of projectile launcher **100** may be located in any suitable position within, or external to, housing **102**.

[0026] Housing **102** may be configured to house various components of projectile launcher **100** that are configured to enable deployment of magazine **104**, provide an electrical current to magazine **104**, and otherwise aid in the operation of projectile launcher **100**, as discussed further herein. Although depicted as a firearm in FIG. **1**, housing **102** may comprise any suitable shape and/or size. Housing **102** may comprise a handle end **112** opposite a deployment end **114**. A deployment end **114** may be configured, sized, and shaped to receive one or more magazine **104** via a magazine receiver **106**. A handle end **112** may be sized and shaped to be held in a hand of a user. For example, a handle end **112** may be shaped as a handle to enable hand-operation of projectile launcher **100** by the user. In various embodiments, a handle end **112** may also comprise contours shaped to fit the hand of a user, for example, an ergonomic grip. A handle end **112** may include a surface coating, such as, for example, a non-slip surface, a grip pad, a rubber texture, and/or the like. As a further example, a handle end **112** may be wrapped in leather, a colored print, and/or any other suitable material, as desired.

[0027] In various embodiments, housing **102** may comprise various mechanical, electronic, and/or electrical components configured to aid in performing the functions of projectile launcher **100**. For example, housing **102** may comprise one or more triggers **108**, control interfaces **110**, processing circuits **202**, user interfaces **204**, power supplies **206**, and/or signal generators **208**. Housing **102** may include a guard (e.g., trigger guard). A guard may define an opening formed in housing **102**. A guard may be located on a center region of housing **102** (e.g., as depicted in FIG. **1**), and/or in any other suitable location on housing **102**. Trigger **108** may be disposed within a guard. A guard may be configured to protect trigger **108** from unintentional physical contact (e.g., an unintentional activation of trigger **108**). A guard may partially and/or fully surround trigger **108** within housing **102**.

[0028] Magazine **104** may comprise and/or be associated with one or more propulsion modules **210** and/or one or more projectiles P. For example, a magazine **104** may comprise and/or be associated with a single propulsion module **210** configured to deploy a single projectile P. As a further example, a magazine **104** may comprise and/or be associated with a single propulsion module **210** configured to deploy a plurality of projectiles P. As an additional example, a magazine **104** may comprise and/or be associated with a plurality of propulsion modules **210** and a plurality of projectiles P, with each propulsion module **210** configured to deploy one or more projectiles P. In various embodiments, and as depicted in FIG. **2**, magazine **104** may be associated with a propulsion module **210** configured to deploy a first projectile P**0**, a second projectile P**1**, a third projectile P**2**, and additional projectile(s) Pn. In additional embodiments, and as depicted in FIGS. **3**A and **3**B, magazine **104** may be associated with a first propulsion module **210-1** configured to deploy a first projectile P**0**, a second propulsion module **210-2** configured to deploy a second projectile P**1**, a third propulsion module **210-3** configured to deploy a third projectile P**2**, and an additional propulsion module **210-**n configured to deploy an additional projectile Pn. Each series of propulsion modules and projectiles may be associated with and/or contained in the same and/or separate magazines.

[0029] In various embodiments, a magazine receiver **106** of housing **102** may be configured to receive and/or couple with one or more magazine **104**. Magazine receiver **106** may be configured as and/or include a channel that secures one or more magazine **104** within and/or to the magazine receiver **106**. Magazine receiver **106** may be configured to receive at least a portion of magazine **104** to passively (i.e., one or more static structures that secure magazine **104**) and/or actively (i.e., one or more dynamic structures that secure magazine **104** by switching from a first state to a second state) secure magazine **104** to housing **102**. For example, magazine receiver **106** may be shaped to comprise an opening in deployment end **114** of housing **102** that permits insertion of magazine **104** within magazine receiver **106**. Alternatively, or in addition, magazine receiver **106** may comprise one or more flanges, rails, ridges, or raised components that guide and/or secure a portion of magazine **104** within magazine receiver **106**. In further examples, magazine receiver may include one or more components that engage magazine **104** based at least in part on magazine **104** being inserted, wherein the one or more components actively secure magazine **104** in response to the insertion. Generally, magazine receiver **106** may comprise one or more mechanical features configured to removably couple one or more magazine **104** within the magazine receiver **106** and in association with projectile launcher **100**. The magazine receiver **106** may be configured to receive a single magazine, two magazines, three magazines, nine magazines, or any other number of magazines.

[0030] In various embodiments, a magazine receiver **106** of housing **102** may be configured as a bay that receives one or more magazine **104**. The bay may comprise an opening at an end of housing **102** sized and shaped to receive one or more magazine **104**. The bay may include one or more mechanical features configured to removably couple one or more magazine **104** within the bay. The bay of housing **102** may be configured to receive a single magazine, two magazines, three magazines, nine magazines, or any other number of magazines.

[0031] In various embodiments, magazine **104** may comprise a magazine interface that is configured to couple with a housing interface associated with housing **102**. Magazine interface and housing interface may be configured to communicate signals, indicators, electrical currents, propellants, and information between magazine **104** and housing **102**. For example, inserting magazine **104** into magazine receiver **106** may enable launching of one or more projectiles P by processing circuit **202**. Additionally, magazine interface and housing interface may comprise one or more devices, sockets, plugs, connectors, nozzles, and other coupling components that enable the communication of signals, substances, and/or information.

[0032] In various embodiments, magazine **104** may comprise a plurality of magazine interfaces and a plurality of housing interfaces associated with housing **102**. Individual magazine interfaces and individual housing interfaces may be configured to communicate at least one of signals, electrical currents, propellants, and information between magazine **104** and housing **102**. For example, inserting magazine **104** into magazine receiver **106** may enable launching one or more projectiles P by processing circuit **202**. The plurality of magazine interfaces and the plurality of housing interfaces may couple when magazine **104** is inserted into housing **102**, forming one or more communication pathways that enable one or more projectiles P to be launched by control circuit **202**.

[0033] Magazine **104** may comprise one or more propulsion modules **210** and one or more projectiles P. For example, a magazine **104** may comprise and/or be associated with a single propulsion module **210** configured to deploy a single projectile P. As a further example, a magazine **104** may comprise and/or be associated with a single propulsion module **210** configured to deploy a plurality of projectiles P. As a further example, a magazine **104** may comprise and/or be associated with a plurality of propulsion modules **210** and a plurality of projectiles P, with each propulsion module **210** configured to deploy one or more projectiles P. In various embodiments, and as depicted in FIGS. **2** and **3**, magazine **104** may be associated with propulsion module **210** configured to deploy a first projectile P**0**, a second projectile P**1**, a third projectile P**2**, and one or more additional projectiles Pn. Alternatively, propulsion module **210** may comprise a plurality of submodules such that a first propulsion submodule is configured to deploy the first projectile P**0**, a second propulsion submodule is configured to deploy the second projectile P**1**, a third propulsion submodule is configured to deploy the third projectile P**2**, and one or more additional propulsion submodules are configured to deploy the one or more addition projectiles Pn. Each series of propulsion modules and projectiles may be associated with and/or contained in the same and/or separate magazines.

[0034] In various embodiments, a propulsion module **210** may be coupled to, or in communication with one or more magazine **104**. In particular, propulsion module **210** may be coupled to, or in communication with magazine **104** such that one or more projectiles P within magazine **104** may be launched, propelled, and/or otherwise driven towards a target. A propulsion module **210** may comprise any device, propellant (e.g., air, gas, etc.), primer, or the like capable of providing a propulsion force in magazine **104** and/or to one or more projectiles P. The propulsion force may include an increase in pressure caused by rapidly expanding gas within an area or chamber. The propulsion force may be directed from the propulsion module to one or more projectiles P in magazine **104** to cause the deployment of the one or more projectiles P. For example, housing **102** and magazine **104** may be in communication via magazine interface and housing interface. Magazine interface and housing interface may be configured to form a connection that provides propulsion force from propulsion module **210** associated with the housing **102** to one or more projectiles P associated with magazine **104**. The propulsion force may be directed from the propulsion module **210**, via the connection formed between magazine interface and housing interface, to cause the deployment of the one or more projectiles P.

[0035] In various embodiments, a propulsion module **210** may be coupled to, or in communication with one or more projectiles P in magazine **104**. In various embodiments, magazine **104** may comprise a plurality of propulsion modules **210**, with each propulsion module **210** coupled to, or in communication with, one or more projectiles P. As noted above, propulsion module **210** may provide a propulsion force in magazine **104**. The propulsion force may include an increase in pressure caused by rapidly expanding gas within an area or chamber that causes deployment of the one or more projectiles P. Similar to examples discussed above, the propulsion force may be directed from the plurality of propulsion modules **210** to one or more projectiles P via one or more connections formed by magazine interface and housing interface.

[0036] In various embodiments, the propulsion force may be directly applied to one or more projectiles P. For example, a propulsion force from propulsion module **210** may be provided directly to first projectile P**0**. A propulsion module **210** may be in fluid communication with one or more projectiles to provide the propulsion force. For example, a propulsion force from propulsion module **210** may travel within a fluid channel associated with housing **102** and magazine **104** to first projectile P**0**. The propulsion force may travel via a manifold in housing **102** and/or magazine **104**.

[0037] In various embodiments, the propulsion force may be provided indirectly to one or more projectiles P. For example, the propulsion force may be provided to a secondary source of propellant within propulsion module **210**. The propulsion force may launch the secondary source of propellant within propulsion module **210**, causing the secondary source of propellant to release propellant. A force associated with the released propellant may in turn provide a force to one or more projectiles P. A force generated by a secondary source of propellant may cause the one or more projectiles P to be deployed from the magazine **104** and projectile launcher **100**.

[0038] In various embodiments, each projectile P**0**, P**1**, P**2**, . . . , and Pn may each be configured to provide a payload to a target. For example, a payload associated with one or more projectiles P may be or include an electrode (e.g., an electrode dart), an entangling projectile (e.g., a tether-based entangling projectile, a net, etc.), a solid substance, a liquid substance, gas substance, or the like. A projectile P may include a spear portion, designed to pierce and/or attach proximate to a tissue of a target in order to provide a conductive electrical path between the electrode and the tissue, as previously discussed herein. Alternatively, or in addition, a projectile P may include a contact portion, designed to slow, mitigate, and/or reduce an impact force of the projectile P on the target.

[0039] In various embodiments, trigger **108** (e.g., projectile launcher trigger, handle trigger, etc.) may be coupled to an outer surface of housing **102**, and may be configured to move, slide, rotate, or otherwise become physically depressed or moved upon application of physical contact. For example, trigger **108** may be actuated by physical contact applied to trigger **108** from within a guard. Trigger **108** may comprise a mechanical or electromechanical switch, button, trigger, or the like. For example, trigger **108** may comprise a switch, a pushbutton, and/or any other suitable type of trigger. Trigger **108** may be mechanically and/or electronically coupled to processing circuit **202**. In response to trigger **108** being activated (e.g., depressed, pushed, etc. by the user), processing circuit **202** may enable deployment of (or cause deployment of) one or more magazine **104** from projectile launcher **100**, as discussed further herein.

[0040] Control interface **110** (e.g., projectile launcher control interface, handle control interface, etc.) of projectile launcher **100** may comprise, or be similar to, any control interface disclosed herein. In various embodiments, control interface **110** may be configured to control selection of firing modes in projectile launcher **100**. Controlling selection of firing modes in projectile launcher **100** may include disabling firing of projectile launcher **100** (e.g., a safety mode, etc.), enabling firing of projectile launcher **100** (e.g., an active mode, a firing mode, an escalation mode, etc.), controlling deployment of magazine **104**, and/or similar operations, as discussed further herein. In various embodiments, control interface **110** may also be configured to perform (or cause performance of) one or more operations that do not include the selection of firing modes. For example, control interface **110** may be configured to enable the selection of operating modes of projectile launcher **100**, selection of options within an operating mode of projectile launcher **100**, or similar selection or scrolling operations, as discussed further herein.

[0041] Control interface **110** may be located in any suitable location on or in housing **102**. For example, control interface **110** may be coupled to an outer surface of housing **102**. Control interface **110** may be coupled to an outer surface of housing **102**, proximate to trigger **108**, and/or a guard of housing **102**. Control interface **110** may be electrically, mechanically, and/or electronically coupled to processing circuit **202**. In various embodiments, in response to control interface **110** comprising electronic properties or components, control interface **110** may be electrically coupled to power supply **206**. Control interface **110** may receive power (e.g., electrical current) from power supply **206** to power the electronic properties or components.

[0042] Control interface **110** may be electronically or mechanically coupled to trigger **108**. For example, and as discussed further herein, control interface **110** may function as a safety mechanism. In response to control interface **110** being set to a “safety mode,” projectile launcher **100** may be unable to launch projectile(s) P from magazine **104**. For example, control interface **110** may provide a signal (e.g., a control signal) to processing circuit **202** instructing processing circuit **202** to disable deployment of projectile(s) P from magazine **104**. As a further example, control interface **110** may electronically or mechanically prohibit trigger **108** from activating (e.g., prevent or disable a user from depressing trigger **108**; prevent trigger **108** from launching a projectile(s) P; etc.).

[0043] Control interface **110** may comprise any suitable electronic or mechanical component capable of enabling selection of firing modes. For example, control interface **110** may comprise a fire mode selector switch, a safety switch, a safety catch, a rotating switch, a selection switch, a selective firing mechanism, and/or any other suitable mechanical control. As a further example, control interface **110** may comprise a slide, such as a handgun slide, a reciprocating slide, or the like. As a further example, control interface **110** may comprise a touch screen, user interface or display, or similar electronic visual component.

[0044] The safety mode may be configured to prohibit deployment of a projectile P from magazine **104** in projectile launcher **100**. For example, in response to a user selecting the safety mode, control interface **110** may transmit a safety mode instruction to processing circuit **202**. In response to receiving the safety mode instruction, processing circuit **202** may prohibit deployment of a projectile P from magazine **104**. Processing circuit **202** may prohibit deployment until a further instruction is received from control interface **110** (e.g., a firing mode instruction). As previously discussed, control interface **110** may also, or alternatively, interact with trigger **108** to prevent activation of trigger **108**. In various embodiments, the safety mode may also be configured to prohibit deployment of a stimulus signal from signal generator **208**, such as, for example, a local delivery.

[0045] The firing mode may be configured to enable deployment of one or more projectiles P from magazine **104** in projectile launcher **100**. For example, and in accordance with various embodiments, in response to a user selecting the firing mode, control interface **110** may transmit a firing mode instruction to processing circuit **202**. In response to receiving the firing mode instruction, processing circuit **202** may enable deployment of a projectile P from magazine **104** by propulsion module **210**. In that regard, in response to trigger **108** being activated, processing circuit **202** may cause the deployment of one or more projectiles P. Processing circuit **202** may enable deployment until a further instruction is received from control interface **110** (e.g., a safety mode instruction). As a further example, and in accordance with various embodiments, in response to a user selecting the firing mode, control interface **110** may also mechanically (or electronically) interact with trigger **108** of projectile launcher **100** to enable activation of trigger **108**.

[0046] In various embodiments, processing circuit **202** may comprise any circuitry, electrical components, electronic components, software, and/or the like configured to perform various operations and functions discussed herein. For example, processing circuit **202** may comprise a processing circuit, a processor, a digital signal processor, a microcontroller, a microprocessor, an application specific integrated circuit (ASIC), a programmable logic device, logic circuitry, state machines, MEMS devices, signal conditioning circuitry, communication circuitry, a computer, a computer-based system, a radio, a network appliance, a data bus, an address bus, and/or any combination thereof. In various embodiments, processing circuit **202** may include passive electronic devices (e.g., resistors, capacitors, inductors, etc.) and/or active electronic devices (e.g., op amps, comparators, analog-to-digital converters, digital-to-analog converters, programmable logic, SRCs, transistors, etc.). In various embodiments, processing circuit **202** may include data buses, output ports, input ports, timers, memory, arithmetic units, and/or the like.

[0047] In various embodiments, processing circuit **202** may include signal conditioning circuitry. Signal conditioning circuitry may include level shifters to change (e.g., increase, decrease) the magnitude of a voltage (e.g., of a signal) before receipt by processing circuit **202** or to shift the magnitude of a voltage provided by processing circuit **202**.

[0048] In various embodiments, processing circuit **202** may be configured to control and/or coordinate operation of some or all aspects of projectile launcher **100**. For example, processing circuit **202** may include (or be in communication with) memory configured to store data, programs, and/or instructions. The memory may comprise a tangible non-transitory computer-readable memory. Instructions stored on the tangible non-transitory memory may allow processing circuit **202** to perform various operations, functions, and/or steps, as described herein.

[0049] In various embodiments, the memory may comprise any hardware, software, and/or database component capable of storing and maintaining data. For example, a memory unit may comprise a database, data structure, memory component, or the like. A memory unit may comprise any suitable non-transitory memory known in the art, such as, an internal memory (e.g., random access memory (RAM), read-only memory (ROM), solid state drive (SSD), etc.), removable memory (e.g., an SD card, an XD card, a CompactFlash card, etc.), or the like.

[0050] Processing circuit **202** may be configured to provide and/or receive electrical signals whether digital and/or analog in form. Processing circuit **202** may provide and/or receive digital information via a data bus using any protocol. Processing circuit **202** may receive information, manipulate the received information, and provide the manipulated information. Processing circuit **202** may store information and retrieve stored information. Information received, stored, and/or manipulated by processing circuit **202** may be used to perform a function, control a function, and/or to perform an operation or execute a stored program.

[0051] Processing circuit **202** may control the operation and/or function of other circuits and/or components of projectile launcher **100**. Processing circuit **202** may receive status information regarding the operation of other components, perform calculations with respect to the status information, and provide commands (e.g., instructions) to one or more other components. Processing circuit **202** may command another component to start operation, continue operation, alter operation, suspend operation, cease operation, or the like. Commands and/or status may be communicated between processing circuit **202** and other circuits and/or components via any type of bus (e.g., SPI bus) including any type of data/address bus.

[0052] In various embodiments, processing circuit **202** may be mechanically and/or electronically coupled to trigger **108**. Processing circuit **202** may be configured to detect an activation, actuation, depression, input, etc. (collectively, an “activation event”) of trigger **108**. In response to detecting the activation event, processing circuit **202** may be configured to perform various operations and/or functions, as discussed further herein. Processing circuit **202** may also include a sensor (e.g., a trigger sensor) attached to trigger **108** and configured to detect an activation event of trigger **108**. The sensor may comprise any suitable sensor, such as a mechanical and/or electronic sensor capable of detecting an activation event in trigger **108** and reporting the activation event to processing circuit **202**.

[0053] In various embodiments, processing circuit **202** may be mechanically and/or electronically coupled to control interface **110**. Processing circuit **202** may be configured to detect an activation, actuation, depression, input, etc. (collectively, a “control event”) of control interface **110**. In response to detecting the control event, processing circuit **202** may be configured to perform various operations and/or functions, as discussed further herein. Processing circuit **202** may also include a sensor (e.g., a control sensor) attached to control interface **110** and configured to detect a control event of control interface **110**. The sensor may comprise any suitable mechanical and/or electronic sensor capable of detecting a control event in control interface **110** and reporting the control event to processing circuit **202**.

[0054] In various embodiments, processing circuit **202** may be electrically and/or electronically coupled to power supply **206**. Processing circuit **202** may receive power from power supply **206**. The power received from power supply **206** may be used by processing circuit **202** to receive signals, process signals, and transmit signals to various other components in projectile launcher **100**. Processing circuit **202** may use power from power supply **206** to detect an activation event of trigger **108**, a control event of control interface **110**, or the like, and generate one or more control signals in response to the detected events. The control signal may be based on the control event and the activation event. The control signal may be an electrical signal.

[0055] In various embodiments, power supply **206** may be configured to provide power to various components of projectile launcher **100**. For example, power supply **206** may provide energy for operating the electronic and/or electrical components (e.g., parts, subsystems, circuits, etc.) of projectile launcher **100** and/or one or more magazine **104**. Power supply **206** may provide electrical power. Providing electrical power may include providing a current at a voltage. Power supply **206** may be electrically coupled to processing circuit **202** and/or signal generator **208**. In various embodiments, in response to a control interface comprising electronic properties and/or components, power supply **206** may be electrically coupled to the control interface **110**. In various embodiments, in response to trigger **108** comprising electronic properties or components, power supply **206** may be electrically coupled to trigger **108**. Power supply **206** may provide an electrical current at a voltage. Electrical power from power supply **206** may be provided as a direct current ("DC"). Electrical power from power supply **206** may be provided as an alternating current ("AC"). Power supply **206** may include a battery. The energy of power supply **206** may be renewable or exhaustible, and/or replaceable. For example, power supply **206** may comprise one or more rechargeable or disposable batteries. In various embodiments, the energy from power supply **206** may be converted from one form (e.g., electrical, magnetic, thermal) to another form to perform the functions of a system.

[0056] Power supply **206** may provide energy for performing the functions of projectile launcher **100**. For example, power supply **206** may provide the electrical current to signal generator **208** that is provided through a target to impede locomotion of the target (e.g., via magazine **104**). Power supply **206** may provide the energy for a stimulus signal. Power supply **206** may provide the energy for other signals, including an ignition signal, as discussed further herein.

[0057] In various embodiments, processing circuit **202** may be electrically and/or electronically coupled to signal generator **208**. Processing circuit **202** may be configured to transmit or provide control signals to signal generator **208** in response to detecting an activation event of trigger **108**. Multiple control signals may be provided from processing circuit **202** to signal generator **208** in series. In response to receiving the control signal, signal generator **208** may be configured to perform various functions and/or operations, as discussed further herein.

[0058] In various embodiments, signal generator **208** may be configured to receive one or more control signals from processing circuit **202**. Signal generator **208** may provide an ignition signal to magazine **104** and/or propulsion module **210** based on the control signals. Signal generator **208** may be electrically and/or electronically coupled to processing circuit **202**, propulsion module **210**, and/or magazine **104**. Signal generator **208** may be electrically coupled to power supply **206**. Signal generator **208** may use power received from power supply **206** to generate an ignition signal. For example, signal generator **208** may receive an electrical signal from power supply **206** that has first current and voltage values. Signal generator **208** may transform the electrical signal into an ignition signal having second current and voltage values. The transformed second current and/or the transformed second voltage values may be different from the first current and/or voltage values. The transformed second current and/or the transformed second voltage values may be the same as the first current and/or voltage values. Signal generator **208** may temporarily store power from power supply **206** and rely at least in part on the stored power to provide the ignition signal. Signal generator **208** may also rely at least in part on received power from power supply **206** to provide the ignition signal, without needing to temporarily store power.

[0059] Signal generator **208** may be controlled at least in part by processing circuit **202**. In various embodiments, signal generator **208** and processing circuit **202** may be separate components (e.g., physically distinct and/or logically discrete). Signal generator **208** and processing circuit **202** may be a single component. For example, a control circuit within housing **102** may at least include signal generator **208** and processing circuit **202**. The control circuit may also include other components and/or arrangements, including those that further integrate corresponding function of these elements into a single component or circuit, as well as those that further separate one or more functions into separate components or circuits.

[0060] Signal generator **208** may be controlled by the control signals to generate an ignition signal having a predetermined current value or values. For example, signal generator **208** may include a current source. The control signal may be received by signal generator **208** to activate the current source at a current value of the current source. An additional control signal may be received to decrease a current of the current source. For example, signal generator **208** may include a pulse width modification circuit coupled between a current source and an output of the control circuit. A second control signal may be received by signal generator **208** to activate the pulse width modification circuit, thereby decreasing a non-zero period of a signal generated by the current source and an overall current of an ignition signal subsequently output by the control circuit. The pulse width modification circuit may be separate from a circuit of the current source or, alternatively, integrated within a circuit of the current source. Various other forms of signal generators **208** may alternatively or additionally be employed, including those that apply a voltage over one or more different resistances to generate signals with different currents. In various embodiments, signal generator **208** may include a high-voltage module configured to deliver an electrical current having a high voltage. In various embodiments, signal generator **208** may include a low-voltage module configured to deliver an electrical current having a lower voltage, such as, for example, 2,000 volts.

[0061] Responsive to receipt of a signal indicating activation of trigger **108** (e.g., an activation event, an activation signal, etc.), processing circuit **202** may provide an ignition signal to magazine **104**, a projectile P in magazine **104**, and/or a propulsion module **210** associated with magazine **104**. For example, signal generator **208** may provide an electrical signal as an ignition signal to magazine **104** in response to receiving a control signal from processing circuit **202**. In various embodiments, the ignition signal may be separate and distinct from a stimulus signal. For example, a stimulus signal in projectile launcher **100** may be provided to a different circuit within magazine **104**, relative to a circuit to which an ignition signal is provided. Signal generator **208** may be configured to generate a stimulus signal. In various embodiments, a second, separate signal generator, component, or circuit (not shown) within housing **102** may be configured to generate the stimulus signal. Signal generator **208** may also provide a ground signal path for magazine **104**, thereby completing a circuit for an electrical signal provided to magazine **104** by signal generator **208**. The ground signal path may also be provided to magazine **104** by other elements in housing **102**, including power supply **206**.

[0062] In various embodiments, projectile launcher **100** may deliver a stimulus signal via a circuit that includes signal generator **208** positioned in the handle and/or handle end **112** of projectile launcher **100**. An interface (e.g., cartridge interface, magazine interface, etc.) mounted on and/or associated with each magazine **104** inserted into housing **102** and/or magazine receiver **106** electrically couples to an interface (e.g., handle interface, housing interface, etc.) in the handle of housing **102**. Signal generator **208** couples to each magazine **104**, and thus to electrodes associated with one or more projectiles P, via the handle interface and the magazine interface. A first filament may couple to magazine interface and to a first electrode associated with a first projectile P**0**. A second filament may couple to the magazine interface and to a second electrode associated with a second projectile P**1**. The stimulus signal travels from signal generator **208**, through the first filament and the first electrode, through target tissue, and through the second electrode and second filament back to signal generator **208**.

[0063] In various embodiments, projectile launcher **100** may further comprise one or more user interfaces **204**. A user interface **204** may be configured to receive an input from a user of projectile launcher **100** and/or transmit an output to the user of projectile launcher **100**. User interface **204** may be located in any suitable location on or in housing **102**. For example, user interface **204** may be coupled to an outer surface of housing **102** or extend at least partially through the outer surface of housing **102**. User interface **204** may be electrically, mechanically, and/or electronically coupled to processing circuit **202**. In various embodiments, in response to user interface **204** comprising electronic or electrical properties or components, user interface **204** may be electrically coupled to power supply **206**. User interface **204** may receive power (e.g., electrical current) from power supply **206** to power the electronic properties or components.

[0064] In various embodiments, user interface **204** may comprise one or more components configured to receive an input from a user. For example, user interface **204** may comprise one or more of an audio capturing module (e.g., microphone) configured to receive an audio input, a visual display (e.g., touchscreen, LCD, LED, etc.) configured to receive a manual input, a mechanical interface (e.g., button, switch, etc.) configured to receive a manual input, and/or the like. In various embodiments, user interface **204** may comprise one or more components configured to transmit or produce an output. For example, user interface **204** may comprise one or more of an audio output module (e.g., audio speaker) configured to output audio, a light-emitting component (e.g., flashlight, laser guide, etc.) configured to output light, a visual display (e.g., touchscreen, LCD, LED, etc.) configured to output a visual, and/or the like.

[0065] In various embodiments, and with reference to FIGS. **3**A and **3**B, a magazine **300** for a projectile launcher is disclosed. Magazine **300** may be similar to any other magazine, deployment unit, or the like disclosed herein (e.g., magazine **104**).

[0066] Magazine **300** may comprise a housing **302** sized and shaped to be inserted into a magazine receiver of a projectile launcher housing, as previously discussed. Housing **302** may comprise a first end **304** (e.g., a deployment end, a front end, etc.) opposite a second end **306** (e.g., a loading end, a rear end, etc.). Magazine **300** may be configured to permit launch of one or more projectiles from first end **304** (e.g., projectiles are launched through first end **304**). Magazine **300** may be configured to permit loading of one or more electrodes from second end **306**. Second end **306** may also be configured to permit provision of stimulus signals from the signal generator of the projectile launcher to the one or more projectiles. In some embodiments, magazine **300** may also be configured to permit loading of one or more electrodes from first end **304**.

[0067] In various embodiments, housing **302** may define one or more bores **308**. A bore **308** may comprise an axial opening through housing **302**, defined and open on first end **304** and/or second end **306**. Each bore **308** may be configured to receive a projectile (or cartridge containing a projectile). Each bore **308** may be sized and shaped accordingly to receive and house a projectile (or cartridge containing a projectile) prior to and during deployment of the projectile from magazine **300**. Each bore **308** may comprise any suitable deployment angle. One or more bores **308** may comprise similar deployment angles. One or more bores **308** may comprise different deployment angles. Housing **302** may comprise any suitable or desired number of bores **308**, such as, for example, two bores, five bores, nine bores, ten bores (e.g., as depicted), and/or the like.

[0068] In various embodiments, housing **302** may be configured to receive one or more cartridges **310**. A cartridge **310** may comprise a body **312** housing a projectile and/or one or more components utilized to deploy the projectile from body **312**. For example, cartridge **310** may comprise a projectile and a propulsion module. The projectile may be similar to any other electrode, projectile, or the like disclosed herein. The propulsion module may be similar to any other propulsion module, primer, or the like disclosed herein.

[0069] In various embodiments, cartridge **310** may comprise a cylindrical outer body **312** defining a hollow inner portion. The hollow inner portion may house a projectile (e.g., an electrode, a spear, filament wire, projectile payload, projectile head, etc.). The hollow inner portion may house a propulsion module configured to deploy the electrode from a first end of the cylindrical outer body **312**. Cartridge **310** may include a piston positioned adjacent to a second end of the projectile. The propulsion module within cartridge **310** may be positioned such that the piston is located between the projectile and the propulsion module. Cartridge **310** may also have a wad positioned substantially adjacent to the piston, where the wad is located between the propulsion module and the piston.

[0070] In various embodiments, a cartridge **310** may comprise a contact **314** on an end of body **312**. Contact **314** may be configured to allow cartridge **310** to receive an electrical signal from a signal generator or processing circuit associated with a projectile launcher. For example, contact **314** may comprise an electrical contact configured to enable the completion of an electrical circuit between cartridge **310** and the signal generator of the projectile launcher. In that regard, contact **314** may be configured to transmit (or provide) a stimulus signal from the projectile launcher to an electrode within the projectile. As a further example, contact **314** may be configured to transmit (or provide) an electrical signal (e.g., an ignition signal) from the signal generator and/or processing circuit of the projectile launcher to a propulsion module within the cartridge **310**. For example, contact **314** may be configured to transmit (or provide) the electrical signal to a conductor of the propulsion module, thereby causing the conductor to heat up and ignite a pyrotechnic material inside the propulsion module. Ignition of the pyrotechnic material may cause the propulsion module to deploy (e.g., directly or indirectly) the electrode from the cartridge **310**.

[0071] In operation, a cartridge **310** may be inserted into a bore **308** of a magazine **300**. The magazine **300** may be inserted into the magazine receiver of the projectile launcher. The projectile launcher may be operated to deploy a projectile from the cartridge **310** in magazine **300**. Magazine **300** may be removed from the bay of the projectile launcher handle. The cartridge **310** (e.g., a used cartridge, a spent cartridge, etc.) may be removed from the bore **308** of magazine **300**. A new cartridge **310** may then be inserted into the same bore **308** of magazine **300** for additional deployments. The number of cartridges **310** that magazine **300** is capable of receiving may be dependent on a number of bores **308** in housing **302**. For example, in response to housing **302** comprising ten bores **308**, magazine **300** may be configured to receive ten cartridges **310** at the same time. As a further example, in response to housing **302** comprising two bores **308**, magazine **300** may be configured to receive two cartridges **310** at the same time.

[0072] In various embodiments, and with reference to FIG. **4**, projectile launcher **100** may be configured to launch a projectile **400**. Projectile **400** may be similar to, or have similar aspects and/or components with, any projectile discussed herein (e.g., one or more projectiles P). Projectile **400** may comprise a projectile housing **402**, a first end **404**, a second end **406**, and an internal channel **408**. The first end **404** of projectile **400** may be configured to receive a launch force from projectile launcher **100**. The second end **406** of projectile **400** may be configured to impact a target after launch and cause projectile payload **410** to be dispersed proximate to the target. Internal channel **408** of projectile **400** may comprise a volume defined by projectile housing **402**. Additionally, internal channel **408** may secure projectile payload **410**, first stopper **412**, primer assembly **414**, second stopper **416**, and impact absorber **418**. It should be noted that any of the one or more components of projectile **400** may be located in any suitable position within, or external to, projectile housing **402**.

[0073] In various embodiments, a projectile housing may be a structural component that is configured to define a projectile and/or contain various internal components of the projectile. A first end of the projectile housing may be referred to and utilized as a projectile base. A second end of the projectile, opposite the first end, may be referred to and utilized as a projectile tip and/or a projectile head. The projectile base may be configured to receive a driving force, a launch force, and/or another force that launches the projectile from a projectile launcher. The projectile tip may be configured to contact a target. Additionally, the projectile housing may be configured to rupture, break, open, and/or otherwise permit a payload of the projectile to disperse and/or be deployed from within the projectile housing in response to at least the projectile impacting a target.

[0074] In various embodiments, projectile housing **402** may be configured to substantially contain internal components of projectile **400**. Projectile housing **402** may be disposed substantially adjacent to first end **404** and second end **406**. Alternatively, or in addition, first end **404** may comprise a first portion of projectile housing **402** and/or second end **406** may comprise a second portion of projectile housing **402**. First end **404** may be configured as a projectile base that receives a force from a propulsion module to launch projectile **400** from projectile launcher **100**. Second end **406** may be configured as a projectile head that contacts a target. The first portion of projectile housing **402** may define and/or be included in first end **404** to receive the force from the propulsion module and direct the force such that projectile **400** is deployed, launched, and/or otherwise accelerated towards the target. The second portion of projectile housing **402** may define, support, and/or be attached to second end **406** to receive an additional force from the second end **406**. Additionally, projectile housing **402** may comprise an internal surface that defines internal channel **408**, internal channel **408** extending at least partially between the first end **404** to the second end **406**. Projectile housing **402** may comprise an external surface that enables projectile **400** to be inserted into at least a magazine and/or cartridge. Additionally, the external surface of projectile housing **402** may enable projectile **400** to be launched from projectile launcher **100**. It should be noted that while various embodiments of projectile **400** and other projectiles being formed as cylinders, projectile **400** and other projectiles may be configured as any three-dimensional shape capable of defining internal channel **408** and being launched from projectile launcher **100**.

[0075] Projectile housing **402** may be configured such that projectile payload **410** may be dispersed and/or otherwise deployed upon impact at a target. For example, projectile housing **402** may be configured to break, shatter, open, rupture, and/or otherwise deploy projectile payload **410**. Projectile housing **402** may be formed from a material selected to contain projectile payload **410** prior to contact with the target. Additionally, projectile housing **402** may deploy, release, and/or otherwise permit dispersal of projectile payload **410** in response to primer assembly **414** activating. For example, primer assembly **414** may comprise pyrotechnic material that is ignited in response to projectile **400** contacting the target and emits gases that deploy projectile payload **410** from within projectile housing **402**. Alternatively, or in addition, primer assembly **414** may comprise a propellant source that emits gases to deploy projectile payload from within projectile housing **402**. It should be noted that primer assembly **414** may be configured to break, shatter, rupture, and/or otherwise open projectile housing **402** to deploy projectile payload **410** and/or cause projectile payload **410** to be dispersed from within projectile housing **402**.

[0076] First end **404** of projectile **400** may be configured as a projectile base that receives the launch force generated by a propulsion module of projectile launcher **100**. The launch force applied to first end **404** may accelerate projectile **400** and cause projectile **400** to launch from projectile launcher **100**. Launching projectile **400** from projectile launcher **100** may cause projectile **400** to travel towards the target. First end **404** may comprise a first diameter D**1** and a first opening associated with internal channel **408**. Additionally, first stopper **412** may be located proximate to and/or within first end **404**. First stopper may be configured to substantially prevent projectile payload **410** from leaking, spilling, and/or otherwise exiting projectile **400** via first end **404**. As noted above, and in various embodiments, first end **404** may include a portion of projectile housing **402** that is configured to secure components of projectile **400** associated with the first end **404**. Further, the portion of projectile housing **402** included in first end **404** may provide some functionality of first end **404**. For example, the portion of projectile housing **402** included in first end **404** may receive at least an amount of the launch force.

[0077] Second end **406** of projectile **400** may be configured as a projectile head and/or a projectile tip that impacts the target after launch. Second end **406** may be further configured to contain and/or secure at least primer assembly **414**, second stopper **416**, and impact absorber **418**. Additionally, second end **406** may comprise a second diameter D**2** and a second opening associated with internal channel **408**. Primer assembly **414** and/or second stopper **416** may be secured by projectile housing **402** proximate to and/or within the second end **406**. Second stopper **416** and/or primer assembly **414** may be configured to substantially prevent projectile payload **410** from leaking spilling and/or otherwise leaving projectile **400** via second end **406**. As noted above, and in various embodiments, second end **404** may include a portion of projectile housing **402** that is configured to secure components of projectile **400** associated with second end **406**. Further, the portion of projectile housing **402** included in second end **406** may provide some functionality of second end **406**. For example, the portion of projectile housing **402** included in second end **406** may receive at least an amount of an applied force received by impact absorber **418** in response to second end **406** contacting the target.

[0078] In various embodiments, projectile housing **402** may define at least a portion of projectile **400**, first end **404**, and/or second end **406**. Projectile housing **402** may include one or more steps, slopes, ridges, and/or other variations in internal and/or external diameter. First diameter D**1** may be greater than, equal to, and/or less than second diameter D**2** based at least on a configuration of a magazine, a cartridge, and/or other portion of projectile launcher **100**. For example, a cartridge associated with projectile **400** may be configured such that first diameter D**1** is less than second diameter D**2** and projectile housing **402** comprises a step where the diameter of projectile housing **402** transitions from first diameter D**1** to second diameter D**2**. Alternatively, projectile housing **402** may comprise a step that changes a thickness of projectile housing **402**. Further, projectile housing **402** may be configured such that first diameter D**1** is substantially equal to second diameter D**2** (as depicted by FIG. **4**).

[0079] In various embodiments, projectile housing **402** may comprise an external housing surface and an internal housing surface. The external housing surface may be configured to be disposed proximate to an internal surface of a bore, firing tube, or other recess associated with a magazine and/or cartridge. Additionally, the external housing surface may permit projectile **400** to be launched from the magazine and/or the cartridge by projectile launcher **100**. The internal housing surface may be configured to secure, couple with, and/or otherwise enable various components of projectile **400** to be disposed within projectile housing **402**.

[0080] First end **404** may comprise first stopper **412**, wherein first stopper **412** may be configured to contain projectile payload **410**. Additionally, first stopper **412** may be configured to bound and/or define an interior of projectile **400**, wherein the interior of projectile **400** contains projectile payload **410** and primer assembly **414**. First stopper **412** may be partially disposed and/or fully disposed within projectile housing **402**. First stopper **412** may be permanently and/or removably coupled to an internal surface of projectile housing **402**. For example, first stopper **412** may be configured to remain substantially static relative to projectile house **402** upon launch of projectile **400** and upon impact of projectile **400** with the target. Alternatively, or in addition, first stopper **412** may be integrated with projectile housing **402**.

[0081] In various embodiments, first stopper **412** may be configured for installation within internal channel **408**. First stopper **412** may share a central axis with internal channel **408** and extend radially out from the central axis to contact an internal surface of projectile housing **402**. First stopper **412** may be coupled with the internal surface by an adhesive, a mechanical component, and/or other coupling component. Alternatively, or in addition, first stopper **412** may be secured in a first stopper position within internal channel **408** by the internal surface of projectile housing **402** by an adhesive, a mechanical component, a size of first stopper **412**, a diameter of internal channel **408**, and/or other configuration of first stopper **412**. Additionally, a substantially fluid-tight seal may be formed between first stopper **412** and the internal surface to substantially prevent the projectile payload from passing between first stopper **412** and the internal surface.

[0082] In various embodiments, first stopper **412** may be configured to modulate, control, and/or otherwise manage acceleration of projectile **400**. For example, primer assembly **414** may be configured such that sudden acceleration and/or deceleration may activate, trigger, and/or otherwise cause primer assembly **414** to disperse projectile payload **410**. First stopper **412** may be configured to manage accelerating forces during launch of projectile **400** from projectile launcher **100** by extending an acceleration period through compression, decompression, and other deformations of first stopper **412**. Extension of the acceleration period may substantially prevent early, inadvertent, and/or accidental triggering of primer assembly **414**.

[0083] In various embodiments, first stopper **412** may be configured as a substantially static structure that does not experience compression and/or deformation during acceleration. First stopper **412** may be configured to remain substantially static such that acceleration of projectile **400** is modulated, controlled, and/or otherwise managed by a propulsion module associated with projectile launcher **100**. Further, primer assembly **414** may be configured such that activation of primer assembly **414** is dependent on at least deceleration of projectile **400** after launch from projectile launcher **100**.

[0084] Second end **406** may comprise primer assembly **414**, second stopper **416**, and impact absorber **418**, wherein primer assembly **414** may be configured to contain projectile payload **410**. Additionally, primer assembly **414**, second stopper **416**, and/or impact absorber **418** may be configured to bound and/or define portions of projectile **400**. Second stopper **416** may be partially disposed and/or fully disposed within projectile housing **402**. Second stopper **416** may be permanently and/or removably coupled to an internal surface of projectile housing **402**. Similarly, impact absorber **418** may be permanently and/or removably coupled to the internal surface of projectile housing **402**. Primer assembly **414** may be transiently coupled and/or associated with the internal surface of projectile housing **402**. For example, primer assembly **414** may be configured to remain substantially static relative to projectile housing **402** upon launch of projectile **400** and translate within projectile housing **402** upon impact with the target. Second stopper **416** may be configured to remain substantially static relative to projectile housing **402** upon launch of projectile **400** and upon impact of projectile **400** with the target. Impact absorber **418** may be configured to remain substantially static upon launch of projectile **400** and to compress upon impact of projectile **400** with the target.

[0085] Internal channel **408** may be defined by the internal surface of projectile housing **402**. Portions of internal channel **408** may be separated and/or bounded by the internal components of projectile **400**. For example, a first portion of internal channel **408** may be defined by a first opening in projectile housing **402** and first stopper **412**. The first portion of internal channel **408** and first stopper **412** may be configured to receive the launch force from a propulsion module associated with projectile launcher **100**. Similarly, a second portion of internal channel **408** may be defined by first stopper **412** and primer assembly **414**, wherein the second portion contains projectile payload **410** prior to impact of projectile **400** upon the target. A third portion of internal channel **408** may be defined by primer assembly **414** and second stopper **416**. The primer assembly **414** may be configured to translate within the third portion upon impact with the target to contact ignition pin **428**, wherein ignition of primer charge **426** may cause expanding gases within at least the third portion to rupture projectile housing **402**. Rupture of projectile housing **402** may further cause projectile payload **410** to be dispersed from the second portion of internal channel **408**.

[0086] Projectile payload **410** may be fluid, powder, and/or other substance that dispersed upon impact of projectile **400** upon the target. Projectile payload **410** may at least partially fill the second portion of the internal channel between first stopper **412** and primer assembly **414**. Additionally, projectile payload **410** may comprise a substance selected for disincentivizing the target from a course of action (e.g., pepper projectiles may disincentivize a person from escalating/continuing a conflict), marking the target for later identification (e.g., an ink may enable vehicles, individuals, and/or property to be identified), and/or otherwise interacting with the target. Further, projectile payload **410** may be substantially contained within internal channel **408** prior to impact upon the target and dispersed from internal channel **408** after impact upon the target.

[0087] In various embodiments, a primer assembly may be a component of a projectile that causes a projectile payload to be deployed proximate to a target of the projectile. The primer assembly may be configured to reside within the projectile and activate upon contact with the target. In response to contacting the target, the primer assembly may cause the projectile payload to be deployed from the projectile such. The primer assembly may be configured as primer assembly **414**. Primer assembly **414** may comprise primer securing ring **420**, primer wall **422**, primer igniter **424**, and primer charge **426**. Primer assembly **414** may be configured to remain substantially static relative to projectile housing **402** and projectile payload **410** during launch of projectile **400** from projectile launcher **100**. Additionally, primer assembly **414** may be configured to activate upon impact of projectile **400** upon the target. For example, primer assembly **414** may be disposed in a rest state, an initial state, an assembled state, and/or other first state when projectile **400** is associated with projectile launcher **100**. Further, primer assembly **414** may be configured to substantially remain the first state prior to impact of projectile **400** upon the target independent of additional handling and/or management of projectile **400**. In response to impact upon the target, primer assembly may transition from the first state to an activated state, a triggered state, and/or other second state. Transition from the first state to the second state may result in primer assembly **414** dispersing projectile payload **410** and/or causing projectile payload **410** to be dispersed by projectile **400**.

[0088] In various embodiments, primer assembly **414** may be configured to activate in response to projectile **400** impacting the target. Primer assembly **414** may be activated by momentum, kinetic energy, decelerating forces, and/or other projectile dynamics associated with a launched projectile **400**. Activation of primer assembly **414** may cause primer assembly **414** to translate within internal channel **408** in response to impact with the target. For example, primer securing ring **420** may be configured to substantially prevent translation of primer assembly **414** by applied forces less than a force threshold. Impact of projectile **400** upon the target may cause projectile **400** and primer assembly **414** to experience a decelerating force. The decelerating force is applied to primer assembly **414** by contact between projectile housing **402** and primer securing ring **420**. Where the decelerating force exceeds the force threshold, primer assembly **414** may translate within internal channel **408**. Translation of primer assembly **414** may cause primer igniter **424** to contact ignition pin **428**, wherein contact between primer igniter **424** and ignition pin **428** ignites primer charge **426**.

[0089] In various embodiments, a primer lock may be configured to secure a primer assembly in place prior to impact of a projectile upon a target. The primer lock may secure the primer assembly via mechanical components, friction, and/or other securing components. Additionally, the primer lock may be configured to release the primer assembly in response to the projectile impacting the target. Further, the primer lock may be configured to secure the primer assembly such that the primer assembly remains substantially static, affixed, attached, and/or other secured in association with the projectile prior to impact upon the target. The primer lock may be a component of the projectile that secures the primer assembly to the projectile. Alternatively, the primer lock may be a component of the primer assembly that secures the primer assembly to the projectile. Alternatively, the primer lock may be a separate component that secures the primer assembly to the projectile.

[0090] In various embodiments, a primer lock may be configured as primer securing ring **420**. Primer securing ring **420** may be configured as a ring that is disposed radially outward from primer wall **422** and radially within projectile housing **402**. Additionally, primer securing ring **420** may be configured to contact the internal surface of projectile housing **402** and the external surface of primer wall **422**. Primer securing ring **420** may space primer wall **422** from projectile housing **402**. Contact with projectile housing **402** and primer wall **422** may enable primer securing ring **420** to substantially prevent translation of primer assembly **414** relative to projectile housing **402** prior to projectile **400** contacting the target. For example, primer securing ring **420** may be formed from a plastic, elastomeric, polymeric, and/or other suitable material for securing primer assembly **414** prior to impact and releasing primer assembly **414** in response to impact upon the target. It should be noted that suitability of material may be associated with primer securing ring **420** being configured to have a force threshold that substantially prevents accidental, uninitiated, and/or otherwise undesired deployment of projectile payload **410** and/or enables release of primer assembly in response to contact with the target.

[0091] Primer securing ring **420** may be associated with a force threshold. For an applied force and/or applied forces less than the force threshold, primer securing ring **420** may substantially prevent translation of primer assembly **414** relative to projectile housing **402**. For an additional applied force and/or additional applied forces greater than the force threshold, primer securing ring **420** may release primer assembly **414** and/or permit primer assembly **414** to translate relative to projectile housing **402**. The force threshold may be determined by the force of friction between at least two of projectile housing **402**, primer assembly **414**, and/or primer securing ring **420**. Similarly, the applied force(s) and/or the additional applied force(s) may be associated with projectile acceleration, projectile deceleration, projectile handling, and/or other forces experienced by projectile **400**. Alternatively, the force threshold may be determined by a shear strength, tensional strength, and/or other material property associated with an adhesive and/or mechanical component coupling primer securing ring **420** to projectile housing **402** and/or primer assembly **414**.

[0092] In various embodiments, a primer wall may be used to contain a dispersal mechanism for a projectile payload. The primer wall may be configured to contain the dispersal mechanism prior to and during translation between a rest position (e.g., a first position, an assembled position, an initial position, etc.) and an activation position (e.g., a second position, an activated position, a dispersal position, etc.). The primer wall may be configured as primer wall **422** and may be configured to contain primer igniter **424** and primer charge **426**. For example, primer wall **422** may be configured as a hollow cylinder that primer igniter **424** and primer charge **426** are disposed within. Primer igniter **424** and/or primer charge **426** may be coupled to, secured by, and/or otherwise contained within primer wall **422**.

[0093] Primer wall **422** may comprise one or more ring stops **430** disposed on the radially outward surface of primer wall **422**. One or more ring stops **430** may be configured as a ridge, raise flange, recess, and/or other mechanical component that substantially prevents disassociation of primer assembly **414** and primer securing ring **420** prior to impact with the target. Ring stop(s) **430** may be further configured such that at least a portion of ring stop **430** extends radially outward beyond a radially internal portion primer securing ring **420**. Additionally, the portion of ring stop **430** may be disposed within internal channel **408** such that translation of primer assembly **414** causes ring stop **430** to contact primer securing ring **420**. Ring stop **430** may be configured to substantially prevent translation of primer assembly **414** for applied forces less than the force threshold associated with primer securing ring **420**. Similarly, ring stop **430** may be configured to cause primer securing ring **420** to translate with primer assembly **414** for applied forces greater than the force threshold. Alternatively, ring stop **430** may be configured to translate past ring stop **430** and permit translation of primer assembly **414** for applied forces greater than the force threshold.

[0094] In various embodiments, a primer activator may be configured to initiate dispersion of the projectile payload in response to the projectile contacting the target. The primer activator may be configured to activate the dispersal mechanism and/or initiate dispersion of the projectile payload in response to proximity with an activator contact, impact with the activator contact, by completion of an activator circuit, and/or other action that causes the dispersal mechanism to disperse the projectile payload. For example, the primer activator may be configured as primer igniter **424** and dispersal mechanism may be configured as primer charge **426**. Additionally, activator contact may be configured as ignition pin **428**. Primer igniter **424** and ignition pin **428** may be configured to cause primer charge **426** to combust and disperse projectile payload **410** from projectile housing

402.
[0095] An activator pin may be a component that is secured within a projectile that is configured to at least partially initiate dispersion of a projectile payload in response to the projectile contacting a target. For example, the activator pin may be configured to contact a dispersion mechanism within the projectile in response to contact with the target. The dispersion mechanism may be ignited, compressed, punctured, ruptured, and/or otherwise activated by contact with the activator pin. Additionally, the activator pin may be configured to remain substantially static relative to the projectile upon impact at the target such that the dispersion mechanism translating within the projectile contacts the activator pin. Alternatively, the activator pin may be configured to translate within the projectile, in response to the projectile impacting the target, to contact the dispersion mechanism. For example, the activator pin may be configured as ignition pin **428** that is contacted by the dispersion mechanism configured as the primer assembly **414** and initiates combustion of primer charge **426**.
[0096] Primer igniter **424** and ignition pin **428** may be configured to collide upon impact of projectile **400** with a target. Primer securing ring **420** may release primer wall **422** in response to impact and permit primer wall **422**, primer igniter **424**, and/or primer charge **426** to translate towards primer pin **428**. Primer igniter **424** may be associated with sufficient energy (e.g., kinetic energy) to contact primer pin **428** and/or compress primer charge **426** between primer igniter **424** and primer pin **428**. Collision of primer igniter **424** and primer pin **428** may ignite primer charge **426**. Alternatively, or in addition, compression of a portion of primer charge **426** by primer igniter **424** and primer pin **428** may ignite primer charge **426**.
[0097] Primer pin **428** may be at least partially secured by second stopper **416** and/or impact absorber **428**. For example, primer pin **428** may extend at least partially through second stopper **416** and/or impact absorber **418**. Primer pin **428**, second stopper **416**, and/or impact absorber **418** may be configured to maintain primer pin **428** in a substantially static position before and after projectile **400** impacts the target. Additionally, second stopper **416** may comprise a pin channel that receives and/or secures primer pin **428**. Further, impact absorber **418** may comprise a pin recess that receives and/or secures primer pin **428**.
[0098] A projectile tip associated with a projectile may be configured to contact a target and decelerate the projectile. Additionally, the projectile tip may be configured to disperse an impact pressure of the projectile to mitigate harm and/or injury of the target. The projectile tip may be configured to compress, radially expand, and/or otherwise deform in response to impacting the target. The projectile tip may be configured as impact absorber **418**, wherein impact absorber **418** is configured to decelerate projectile **400** and cause primer assembly **414** to activate.
[0099] In various embodiments, impact absorber **418** may be coupled to an end of housing **402**. In various embodiments, impact absorber **418** may be partially received in housing channel **408**. Impact absorber **418** may comprise a first portion disposed at least partially within projectile housing **402** and a second portion coupled to the first portion that is disposed external to housing **402**. The second portion of impact absorber **418** may be configured to contact the target and receive the applied force that decelerates projectile **400**. In various embodiments, the first portion of impact absorber **418**, the second portion of impact absorber **418**, and/or projectile housing **402** may be formed as and/or integrated into a single component. In various embodiments, a diameter of impact absorber **418** at a second end **406** of projectile **400** may be greater than a first diameter D**1** of first end **404**. In various embodiments, impact absorber **418** may be configured to compress, spread, and/or otherwise deform to mitigate the applied force of impact at the target. In particular, a travel velocity of projectile **400** may be decelerated over a duration that may be managed via the impact absorber **418**. The impact absorber **418** may be configured to substantially resist deformation such that the deceleration of projectile **400** is accomplished over a short duration. The impact absorber **418** may be configured to deform such that the deceleration of projectile **400** is accomplished over a long duration. The duration length may be managed through the deformation ability of impact

absorber **418** and may result in impact absorber **418** being configured such that the applied force satisfies the force threshold associated with primer assembly **414** and activates primer assembly **414**.

[0100] In various embodiments, projectile **400** may comprise a plurality of portions that are combined to disperse the projectile payload in proximity to the target. For example, a first portion **432** of projectile **400** may comprise a projectile base that receives the launch force from the projectile launcher. Additionally, a second portion **434** of projectile **400** may comprise projectile payload **410** and/or other projectile payload contained by projectile **400**. Further, a third portion **436** of projectile **400** may comprise a primer assembly **414** that is configured to activate based at least in part on impact absorber **408** contacting the target. A fourth portion **438** of projectile **400** may comprise a projectile tip that contacts the target and transfers an applied force to projectile **400**. It should be noted that the plurality of portions may comprise duplicates of individual portions discussed above and one or more additional portions with other functionality.

[0101] In various embodiments, the first portion **432** (e.g., projectile base portion, base portion, etc.) may reference and/or comprise one or more components of projectile **400** discussed above. First portion **432** may comprise the projectile base, first end **404**, first stopper **412**, and other components of projectile **400**. First portion **432** may be configured as a section of projectile **400** that receives the launch force and causes projectile **400** to be accelerated to a travel velocity towards the target. Additionally, and upon impact at the target, first portion **432** may be configured to at least partially direct the dispersal of projectile payload **410** and/or substantially preventing projectile payload **410** from exiting projectile **400** via first end **404**.

[0102] In various embodiments, the second portion **434** (e.g., projectile payload portion, payload portion, storage portion, etc.) may reference and/or comprise one or more components of projectile **400** discussed above. Second portion **434** may comprise projectile payload **410**, first stopper **412**, an additional stopper, one or more faults associated with projectile housing **402**, and/or other components for dispersing projectile payload **410**. Second portion **434** may be configured as a section of projectile **400** that enables primer assembly **414** to disperse projectile payload **410** from projectile **400**. Additionally, second portion **434** may be configured to substantially prevent projectile payload **410** from being dispersed prior to the projectile tip contacting the target. Further, second portion **434** may comprise one or more faults that are configured to enable dispersal of projectile payload **410** based at least in part on the one or more faults causing projectile housing **402** to rupture.

[0103] In various embodiments, second portion **434** may be configured to include a segment of projectile housing **402**. For example, second portion **434** may comprise a segment of projectile housing **402** that is configured to contain projectile payload **410**, preserve projectile payload **410**, disperse projectile payload **410**, and/or otherwise secure projectile payload **410** prior to contact with a target. Second portion **434** may be associated with the segment of projectile housing **402** that is configured to contain projectile payload **410**. For example, projectile housing **402** may be formed to contain projectile payload **410** and substantially prevent degradation of projectile payload **410** prior to dispersal (e.g., fluid resistant material where projectile payload **410** is a fluid, substantially fluid tight seal between the segment of projectile housing **402** and a stopper where projectile payload **410** is a fluid, nonreactive material where projectile payload may be degraded if exposed to reactive materials, etc.). Additionally, the segment of projectile housing **402** may be formed to preserve projectile payload **410** during storage of projectile **400** and prior to dispersal proximate to the target (e.g., second portion **434** may be formed to substantially prevent projectile payload **410** from contacting air and substantially prevent oxidation of projectile payload **410**). Further, the segment of projectile housing **402** may be formed to include one or more faults that enable dispersal of projectile payload from within second portion **434**.

[0104] In various embodiments, the third portion **436** (e.g., primer assembly portion, primer portion, dispersing portion, etc.) may reference and/or comprise one or more components of projectile **400** discussed above. Third portion **436** may comprise primer assembly **414**, second stopper **416**, ignition pin **428**, and/or other components for dispersing projectile payload **410**. Third portion **436** may be configured as a section of projectile **400** that comprises primer assembly **414** and activates primer assembly **414** in response to the projectile tip contacting the target. Additionally, third portion **436** may be configured to substantially prevent primer assembly **414** from activating prior to the projectile tip contacting the target.

[0105] In various embodiments, third portion **436** may be configured to include a segment of projectile housing **402**. The segment of projectile housing **402** may be configured to enable functionality of third portion **436** and/or may be shared with second portion **434**. For example, third portion **436** may comprise a segment of projectile housing **402** that is configured to secure primer assembly **414** prior to contact with a target, enable activation of primer assembly **414** based at least in part on contacting the target, enable primer assembly **414** to disperse projectile payload **410** from within second portion **434** proximate to the target, and/or otherwise cause projectile payload **410** to be dispersed from within projectile **400**. Third portion **436** may comprise a segment of projectile housing **402** that is configured to secure primer assembly **414** via primer securing ring **420**. For example, the segment of projectile housing **402** may be configured to couple with primer securing ring **420** such that primer assembly **414** is substantially secured in a first position prior to contact with the target. Alternatively, or in addition, the segment of projectile housing **402** may comprise a channel that at least a portion of primer securing ring **420** extends within to substantially secure primer assembly in the first position. Further, the segment of projectile housing **402** may be configured to decouple and/or otherwise permit translation of primer securing ring **420** based at least on projectile **400** contacting the target. Similar to second portion **434**, third portion **436** may be associated with a segment of projectile housing **402** that comprises one or more faults that enable projectile payload **410** to be dispersed from second segment **434**.

[0106] In various embodiments, the fourth portion **438** (e.g., projectile tip portion, impact portion, contact portion, etc.) may reference and/or comprise one or more components of projectile **400** discussed above. Fourth portion **438** may comprise the projectile tip, second stopper **416**, impact absorber **418**, and/or other components of projectile **400**. Fourth portion **438** may be configured as a projectile tip of projectile **400** that contacts the target and transfers an applied force to the projectile **400**.

[0107] In various embodiments, the plurality of portions may be configured for functions that operate sequentially and/or in parallel to disperse projectile payload **410** in proximity to the target. Additionally, individual components of projectile **400** may be utilized by one or more portions of projectile **400** during dispersal of projectile payload **410**. For example, second stopper **416** may be configured to secure ignition pin **428** and impact absorber **418**, as depicted by FIG. **4**. Second stopper **416** may be considered a component within third portion **436** as ignition pin **428** is contacted by and activates primer assembly **414**. Additionally, second stopper **416** may be considered a component within fourth portion **438** as second stopper **416** is substantially secured by projectile housing **402**. In response to impact absorber **418** contacting the target, second stopper **416** may receive at least a portion of the applied force of impact at the target that decelerates projectile **400**. Accordingly, the plurality of portions may overlap such that individual components of projectile **400** are included in one or more portions. Alternatively, or in addition, individual portions may comprise duplicate components (e.g., second stopper **416** is configured as two stoppers so that a single hopper is associated with the third portion **436** and the fourth portion **438**).

[0108] In various embodiments, the plurality of portions may be arranged to disperse projectile payload **410** from within projectile housing **402**. Generally, the first portion **432** and the fourth portion **438** will be disposed at first end **404** and second end **406**, respectively. Additionally, one or more second portions **434** and/or one or more third portions **436** may be included in an example arrangement. For example, a first end of an example projectile may be configured as a first portion **432**. A second portion **434** may be disposed substantially adjacent to the first portion **432**. A third portion **436** may be disposed substantially adjacent to the second portion **434**. An additional second portion **434** may be disposed substantially adjacent to the third portion **436** opposite the second portion **434**. A fourth portion **438** may be disposed adjacent to the additional second portion **434** substantially opposite the third portion **436** and be configured as the second end of the example projectile. Alternatively, the plurality of portions may be arranged as depicted by FIG. **4**. Additionally, the plurality of portions may be arranged various combinations that enable primer assembly **414** to disperse projectile payload **410** from within projectile housing **402**. Further, the plurality of portions may be arranged substantially coaxially to form the projectile **400** and/or other example projectile.

[0109] In various embodiments, and with reference to FIGS. **5**A-**5**C, a projectile launcher **100** may be configured to launch projectile **400** at a target. Projectile **400** may be similar to, or have similar aspects and/or components with, any propulsion module discussed herein (e.g., one or more projectiles P, projectile **400** as depicted by FIG. **4**, etc.). Projectile **400** may comprise a projectile housing **402**, a first end **404**, a second end **406**, and an internal channel **408** that are substantially similar to those described with reference to FIG. **4**. Similarly, primer assembly **414** may comprise primer securing ring **420**, primer wall **422**, primer igniter **424**, and primer charge **426** that are substantially similar to those described with reference to FIG. **4**.

[0110] It should be noted that projectile **400** may be associated with a series of projectile states based at least in part on whether projectile **400** has contacted the target. For example, a first state of projectile **400**, as depicted by FIG. **5**A, may be referenced as a rest state, a flight state, a launch state, an inactive state, an inert state, and/or other state where internal components of projectile **400** remain substantially static. Additionally, a second state of projectile **400**, as depicted by FIG. **5**B, may be referenced as an impact state, a transition state, an activating state, and/or other state where primer assembly **414** initiates dispersion of projectile payload **410**. Further, a third state of projectile **400**, as depicted by FIG. **5**C, may be referenced as a dispersion state, a deployed state, an activated state, and/or other state where projectile payload **410** is dispersed in proximity to the target.

[0111] In the first state, projectile **400** may be disposed within a magazine of projectile launcher **100**, within a cartridge, in flight to a target, in the process of being launched, and/or otherwise disassociated with a target surface **502**. Additionally, the first state may be associated with internal housing surface **504** contacting external ring surface **506** and securing primer securing ring **420** relative to projectile housing **402**. Securing primer securing ring **420** relative to projectile housing **402** may comprise primer assembly **414** remaining substantially static in position within projectile housing **402**. Contact between internal housing surface **504** and external ring surface **506** may be further associated with a contact force that results from the primer securing ring **420** being disposed between projectile housing **402** and primer wall **422**. Primer securing ring **420** may separate primer assembly **414** from an inner surface of projectile housing **402**. Primer securing ring **420** may establish a spacing around an external circumference of primer assembly **414** and projectile housing **402**. Additionally, primer securing ring **420** may be configured to fit within one or more channels associated with projectile housing **402** (e.g., a channel associated with the inner surface of projectile housing), primer assembly **414** (e.g., a channel associated with the external surface of primer wall **422**), and/or other internal component of projectile **400**. At least one of the external ring surface **506** and the internal ring surface **514** may be configured to permit translation of primer assembly **414** based at least on the force threshold being satisfied by an applied force (e.g., impact absorber **418** contacting the target). Further, an axial thickness of primer securing ring **420** (e.g., a thickness of primer securing ring **420** substantially parallel to a central axis of projectile **400**) may be configured to provide a force threshold. For example, the axial thickness may be utilized to manage an amount of force that satisfies the force threshold and deform primer securing ring **420** to enable primer assembly to translate within third portion **436**.

[0112] Primer securing ring **420** may be configured to have a rest thickness (e.g., a distance between external ring surface **506** and an internal ring surface **514** while primer securing ring **420** does not experience compressive force(s)) that is compressed when placed between projectile housing **402** and primer wall **422**. Compression of primer securing ring **420** may be used to determine and/or establish a force threshold associated with activation of the primer assembly **414**. For example, compression of primer securing ring **420** may result in the application of a contact force between at least internal housing surface **504** contacting external ring surface **506**. The contact force may function as at least a portion of a normal force (e.g., a force perpendicular to internal housing surface **504** contacting external ring surface **506** that results from the two surfaces being pressed together). The contact force may correlate with the force threshold for initiating translation of primer assembly **414**.

[0113] In various embodiments, the contact force may result in the force threshold comprising at least a static frictional force that is overcome to permit translation of primer assembly **414**. In response to an applied force exceeding the static frictional force, and/or any additional forces securing primer assembly **414**, primer assembly **414** may initiate translation within internal channel **408**. Additionally, the applied force may be an external force that is applied to an object (e.g., an accelerating force or decelerating force) or an innate force that is associated with the object (e.g., inertia causing an object to resist changes in motion of the object). It should be noted that a decelerating force (e.g., impact absorber **418** causing projectile **400** to stop upon contacting target **502**) applied by projectile housing **402** to primer assembly **414** may cause inertia of primer assembly **414** to overcome the force threshold and initiate translation of primer assembly **414** within internal channel **408**.

[0114] In various embodiments, the first state may be associated with a first surface **508** of primer igniter **424** at least partially bounding projectile payload **410**. First surface **508** may be a base of primer igniter **424**. Additionally, projectile payload **410** may be bounded by at least first surface **508** and first stopper **412**. In the first state, projectile payload **410** may substantially fill the portion of internal channel **408** between first surface **508** and first stopper **412**. For example, projectile payload **410** may be a powder that is disposed, placed, packed, and/or otherwise contained by at least first surface **508** and first stopper **412**. Further, first surface **508** and first stopper **412** may be configured to substantially prevent shifting of projectile payload **410** during flight of projectile **400**, uneven dispersal of projectile payload **410**, uneven distribution of projectile payload around a central axis A, and/or other inconsistencies that may impact performance of projectile **400**.

[0115] In various embodiments, the first state may be associated with ring face **510** of primer securing ring **420** contacting and/or being disposed proximate to stop face **512** of ring stop **430**. Ring face **510** may be configured to substantially prevent stop face **512** from translating towards second stopper **416**. Additionally, ring face **510** may be configured to permit stop face **512** translated towards second stopper **416** under an applied force greater than the force threshold associated with the primer securing ring **420**. For example, the force threshold may be associated with an applied force capable of causing ring face **510** and primer securing ring **420** to compress, break, deform, and/or otherwise permit stop face **512** and primer wall **422** to translate towards second stopper **416**.

[0116] In various embodiments, stop face **512** may be configured to cause deformation of ring face **510** and primer securing ring **420**. Stop face **512** may comprise a sloped face that applies the applied force to ring face **510**, the applied force causing primer securing ring **420** to compress and/or deform. Where the applied force exceeds the force threshold, primer securing ring **420** may compress and/or deform to permit ring stop **430** to pass radially within internal ring surface **514**. Alternatively, or in addition, the applied force may cause primer securing ring **420** to break, rupture, tear, and/or otherwise permanently deform. Independent of whether primer securing ring **420** compresses, deforms, permanently deforms, and/or otherwise is modified by the applied force, primer securing ring **420** may release primer assembly **414** in response the applied force causing stop face **512** translating into and/or past ring face **510**.

[0117] In various embodiments, projectile **400** may transition from the first state illustrated by FIG. 5A to the second state illustrated by FIG. 5B in response to impact absorber **418** contacting target **502**. Impact absorber **418** may contact target **502** and substantially arrest projectile **400** (e.g., substantially stopping the forward movement of projectile **400**), wherein arresting projectile **400** comprises impact absorber **418** to apply a decelerating force to primer assembly **414**. Impact absorber **418** may apply the decelerating force substantially parallel to central axis A from the second end **406** towards the first end **404**. The decelerating force applied by impact absorber **418** may cause various components of projectile **400** to cease movement towards target **502**. For example, the decelerating force applied by impact absorber **418** may substantially arrest at least projectile housing **402**, first stopper **412**, second stopper **416**, and ignition pin **428**.

[0118] In various embodiments, the decelerating force applied by impact absorber **418** may initiate translation of primer assembly **414** within internal channel **408**. For example, and based at least on forward momentum and/or inertia of primer assembly **414** after launch of projectile **400**, the decelerating force may cause external wall surface **516** to translate past internal ring surface **514**. As noted above, an applied force (e.g., the decelerating force) exceeding the force threshold associated with primer securing ring **420** may cause ring face **510** to contact stop face **512**, cause primer securing ring **420** to be compressed, and permit external wall surface **516** to translate past internal ring surface **514** towards second stopper **416**. Additionally, the force threshold may be associated with at least a static frictional force and a compressive force that are overcome to permit translation of primer assembly **414**.

[0119] In various embodiments, the decelerating force applied by impact absorber **418** may initiate translation of primer assembly **414** within internal channel **408**. For example, and based at least on forward momentum of primer assembly **414** after launch of projectile **400**, the decelerating force may cause external ring surface **506** to translate past internal housing surface **504**. As noted above, an applied force (e.g., the decelerating force) exceeding the force threshold associated with primer securing ring **420** may cause primer assembly **414** to translate towards second stopper **416**. Contact between ring face **510** and stop face **512** may cause external ring surface **506** to translate past internal housing surface **504** towards second stopper **416**. Additionally, the force threshold may be associated with at least a static frictional force that is overcome to permit translation of primer assembly **414**.

[0120] In response to the force threshold being overcome, primer assembly **414** (optionally with primer securing ring **420**) may translate towards and contact ignition pin **428**. Contact surface **518** of primer igniter **424** may be aligned with ignition pin **428** such that translation of primer assembly **414** causes contact surface **518** to contact ignition pin **428**. Additionally, contact surface **518** and ignition pin **428** may be disposed within projectile housing **402** at and/or parallel to central axis A. Alignment to and/or parallel to central axis A may enable translation of primer assembly **414** to impact ignition pin **428**. Further, contact surface **518** and ignition pin **428** may be configured to initiate combustion of primer charge **426**, wherein combustion is initiated by contact surface **518** impacting ignition pin **428**.

[0121] In various embodiments, FIG. 5B illustrates projectile **400** impacting target **502**. As noted above, impact with target **502** causes impact absorber **418** to substantially arrest projectile **400**, wherein deceleration of projectile **400** causes primer assembly **414** to initiate translation. Primer assembly **414**, primer securing ring **420**, and/or projectile housing **402** may be configured such that deceleration of projectile **400** causes primer assembly **414** to translate towards ignition pin **428**. Translation of primer assembly **414** may be associated with sufficient kinetic energy that collision of contact surface **518** with ignition pin **428** activates primer charge **426** (e.g., ignites primer charge **426**). More specifically, primer assembly **414** may be associated with sufficient kinetic energy to overcome an activation energy of primer charge **426**. The activation energy may be associated with ignition of combustible material, puncturing an external wall of a cannister, compression of a pressure ignited material, and/or otherwise activating primer charge **426**.

[0122] In various embodiments, projectile **400** may transition from the second state illustrated by FIG. **5**B to the third state illustrated by FIG. **5**C in response to contact surface **518** of primer assembly **414** colliding with and/or compressing primer charge **426** against ignition pin **428**. For example, activation of primer charge **426** may initiate dispersion of projectile payload **410** from within projectile housing **402**. The transition from the second state to the third state may be characterized by projectile housing **402** being opened and projectile payload **410** being released proximate to target **502**. Additionally, opening of projectile housing **402** may include projectile housing **402** being separated into a plurality of projectile housing portions (e.g., projectile housing portion **520-1**, projectile housing portion **520-2**, projectile housing portion **520-3**, projectile housing portion **520-***n*, etc.). Further, release of projectile payload **410** may result in a dispersed payload **522** proximate to target **502**.
[0123] In various embodiments, a dispersion mechanism may be configured to disperse a projectile payload proximate to a target in response to a projectile contacting the target. For example, the dispersion mechanism may utilize chemical reactions, compressed fluids, mechanical drives, and/or other mechanisms to open (e.g., break, rupture, burst, shatter, etc.) a projectile housing and disperse the projectile payload. Additionally, the dispersion mechanism may activate in response to a collision, impact, and/or other contact with an activator pin and/or other activation device. The activation device may initiate the chemical reaction (e.g., ignites a combustible material or a pyrotechnic material such that expanding gas(es) disperse the projectile payload), puncture a container of pressurized fluid (e.g., punctures a compressed air cannister so that compressed air disperses the projectile payload), and/or otherwise cause the projectile payload to be dispersed.
[0124] Primer assembly **414** may be configured to initiate dispersion of projectile payload **410** via primer charge **426** igniting and outputting an amount of expanding gas(es) sufficient to rupture projectile housing **402**. The ignition of primer charge **426** may release an amount of expanding gases, thermal energy, and/or other force capable of opening projectile housing **402**. Additionally, ignition of primer charge **426** may cause the amount of expanding gases, thermal energy, and/or other force to disperse projectile payload **410** into dispersed payload **522**. Alternatively, or in addition, dispersion of projectile payload **410** into dispersed payload **522** may be achieved by opening of projectile housing **402** via primer charge **426** in proximity to target **502**. For example, opening of projectile housing **402** may cause powders, liquids, gases, and/or other payloads to splash, spread, spill, diffuse and/or otherwise disperse proximate to target **502**.
[0125] Projectile housing **402** may be configured to separate into a plurality of projectile housing portions **520**. For example, projectile housing **402** may be formed of a material that shatters, breaks, and/or otherwise separates into projectile housing portions **520-1** through **520-***n* in response to primer charge **426** activating. Activation of primer charge **426** may at least increase an internal pressure of projectile **400** from a rest pressure (e.g., substantially atmospheric pressure) to an activated pressure, wherein the activated pressure is greater than a pressure threshold associated with projectile housing **402**. In response to the activated pressure exceeding the pressure threshold, projectile housing **402** may separate into the plurality of projectile housing portions **520-1** through **520-***n*.
[0126] In various embodiments, projectile housing **402** may be configured to separate into at least first housing portion **520-1**, second housing portion **520-2**, third housing portion **520-3**, and/or additional housing portion **520-***n*. Projectile housing **402** may be formed, etched, scored, and/or otherwise modified such that first housing portion **520-1**, second housing portion **520-2**, third housing portion **520-3**, and/or additional housing portion **520-***n* are separated by faults. Activation of primer assembly **414** causes projectile housing **402** to separate into the plurality of projectile housing portions **520-1** through **520-***n*, wherein projectile housing **402** opens along the faults. Further, primer charge **426** may be configured such that the activated pressure generated by primer charge **426** is sufficient to cause projectile housing **402** to open along the faults. It should be noted that faults may reference to weakened portions of projectile housing **402** created by formation,

etching, scoring, and/or modification of projectile housing **402**.

[0127] In various embodiments, projectile housing **402** may be configured to separate into at least first housing portion **520-1**, second housing portion **520-2**, third housing portion **520-3**, and/or additional housing portion **520**-*n*. Separation of projectile housing **402** into the plurality of housing portions **520-1** through **520**-*n* and the number of housing portions formed may be at least partially dependent on primer charge **426**. For example, primer charge **426** may generate the activated pressure sufficient to open projectile housing **402** and separate projectile housing **402** into the plurality of housing portions **520-1** through **520**-*n*. Additionally, the activated pressure may form the plurality of housing portions **520-1** through **520-**n at random (e.g., activated pressure rupturing and breaking projectile housing **402**) and/or at pseudorandom (e.g., ruptures and breaks in projectile housing **402** tend to occur proximate to primer assembly **414**). Projectile housing **402** may separate into the plurality of housing portions **520-1** through **520**-*n* in response to distributed weak points associated with projectile housing **402** that occur during formation of projectile housing **402**.

[0128] Primer assembly **414** may be configured to disperse projectile payload **410** into dispersed payload **522**. As noted above, activation of primer charge **426** may cause an amount of expanding gases, thermal energy, and/or other force to be generated by primer assembly **414**. The force generated by primer charge **426** may cause projectile payload **410** to disperse into dispersed payload **522** in addition to opening projectile housing **402**. Additionally, ignition of primer charge **426** may cause the amount of expanding gases, thermal energy, and/or other force to disperse projectile payload **410** into dispersed payload **522**. Alternatively, or in addition, dispersion of projectile payload **410** into dispersed payload **522** may be achieved by opening of projectile housing **402** via primer charge **426** in proximity to target **502**. For example, opening of projectile housing **402** may cause powders, liquids, gases, and/or other payloads to splash, spread, spill, diffuse and/or otherwise disperse proximate to target **502**.

[0129] In various embodiments, projectile **400** may be launched from projectile launcher **100** towards target **502** in the first state depicted by FIG. **5**A. The launch force may be applied by projectile launcher **100** to at least first end **406** of projectile **400** and/or first stopper **412**. The launch force may accelerate projectile **400** to a flight velocity. It should be noted that the launch force may be applied to at least one of projectile housing **402**, first end **406**, and/or first stopper **412** via a mechanical component (e.g., a puncher, piston, and/or other mechanical drive), an expanding fluid (e.g., compressed gas being released, chemical reaction emitting expanding gases, etc.), and/or other launch components.

[0130] In various embodiments, and as depicted by FIG. **5**A, the internal components of projectile **400** may remain substantially static within internal channel **408**. Primer securing ring **420** may remain static relative to projectile housing **402** based at least in part on contact between internal housing surface **504** and external ring surface **506**. Similarly, primer assembly **414** may remain static relative to projectile housing **402** based at least in part on contact between ring face **510** and stop face **512**. Additionally, primer assembly **414** may remain static relative to projectile housing based at least on part on contact between first surface **508** and projectile payload **410** (e.g., primer assembly is substantially prevented from moving toward first stopper **412** by projectile payload **410**).

[0131] In various embodiments, and as depicted by FIG. **5**B, projectile **400** may contact target **502** after launch by projectile launcher **100**. Contact with target **502** may substantially arrest movement of projectile housing **402**, first stopper **412**, second stopper **416**, impact absorber **418**, and ignition pin **428**. Additionally, primer securing ring **420** may compress, break, stretch, and/or otherwise deform such that primer assembly **414** is released. Primer assembly **414** may translate within internal channel such that primer igniter **424** and/or primer charger **426** collide with ignition pin **428**, wherein collision with ignition pin **428** ignites and/or otherwise activates primer charger **426**. Alternatively, primer securing ring **420** may translate within internal channel **408** with primer assembly **414**, wherein ring external surface **506** translates substantially adjacent to internal housing surface **504** to permit translation of primer assembly **414**.

[0132] In various embodiments, an as depicted by FIG. 5C, projectile payload **410** may be released and/or dispersed from within projectile **400**. Activation of primer charge **426** may cause projectile housing **402** to open, releasing and/or dispersing projectile payload proximate to target **502**. Alternatively, or in addition, activation of primer charge **426** may cause first stopper **412** to exit first end **404** and second stopper **416** to exit second end **406**, wherein first stopper **412** and second stopper **416** exiting internal channel **408** may cause projectile payload **410** to be released and/or dispersed proximate to target **502**. Accordingly, projectile **400** may be launched by projectile launcher **100**, impact target **502**, and release projectile payload **410** to create dispersed payload **522** proximate to target **502**.

[0133] In various embodiments, and with reference to FIG. **6**, projectile launcher **100** may be configured to launch a projectile **600**. Projectile **600** may be similar to, or have similar aspects and/or components with, any projectile discussed herein (e.g., one or more projectiles P, projectile **400**, etc.). It should be noted that various components of projectile **600** may also be similar to, or have similar aspects and/or subcomponents with, the various components of projectile **400**. For example, and similar to projectile **400**, projectile **600** may comprise a projectile housing, a first end, a second end, and an internal channel. The first end of projectile **600** may be configured to receive a launch force from projectile launcher **100**. The second end of projectile **600** may be configured to impact a target after launch and cause projectile payload **620** to be dispersed proximate to the target. Additionally, projectile housing **602** may secure first stopper **604**, primer assembly **608**, activation pin **616**, second stopper **618**, and/or impact absorber **622**. It should be noted that any of the one or more components of projectile **600** may be located in any suitable position within, or external to, projectile housing **602**. Primer assembly **608** may be similar to, or have similar aspects and/or components with, primer assembly **414**. Primer assembly **608** may be associated with primer securing ring **606** and comprise primer wall **610**, primer activator **612**, and primer charge **614**.

[0134] A first end of projectile **600** may contain first stopper **604**, primer securing ring **606**, primer assembly **608**, activation pin **616**, and second stopper **618**. First stopper **604** may be permanently and/or removably coupled to an internal surface of projectile housing **602** to substantially prevent a launch force from activating primer assembly **608**. For example, first stopper **604** may be configured to remain substantially static relative to projectile housing **602** upon launch of projectile **600** from projectile launcher **100**. Alternatively, or in addition, first stopper **604** may be integrated with projectile housing **602**. Further, first stopper **604** may be coupled with projectile housing **602** to form a substantially fluid-tight seal.

[0135] The first end of projectile **600** may be configured such that activation of primer assembly **608** causes projectile payload **620** to be deployed from projectile **600**. For example, and in response to projectile **600** impacting a target, primer assembly **608** may translate within projectile housing **602** and collide with activation pin **616**. Primer activator **612** may collide with activator pin **616** and cause primer charge **614** to deploy and/or disperse projectile payload **620** from within projectile housing **602**. In some embodiments, first stopper **604** and second stopper **618** may be secured within projectile housing **602** such that projectile housing **602** opens to permit deployment of projectile payload **620**. Alternatively, or in addition, first stopper **604** may be secured within projectile housing **602** such that at least one of activator pin **616** and/or second stopper **618** is released, dislodged, moved, and/or decoupled within projectile housing **602** to at least partially cause projectile payload **620** to be deployed and/or dispersed.

[0136] In various embodiments, second stopper **618** may be coupled with, attached to, and/or otherwise secured by projectile housing **602**. Second stopper **618** may be secured by projectile housing **602** such that primer assembly **608** may translate within projectile housing **602** and collide with activator pin such that primer charge **614** is activated. Primer charge **614** may be associated with an activation threshold that is associated with an amount of energy for activating primer charge **614**, the amount of energy being provided by primer activator **612** contacting activator pin **616**, primer activator **612** compressing primer charge **614** against activator pin **616**, and/or otherwise applying kinetic energy from primer assembly **608** to primer charge **614**. Second stopper **618** may be configured to remain secured by projectile housing **602** and enable primer charge **614** to be activated by collision of primer assembly **608** and activator pin **616**. It should be noted that second stopper **618** may be configured to remain substantially static under initial contact of primer assembly **608** and activator pin **616**.

[0137] In various embodiments, second stopper **618** may be coupled with, attached to, and/or otherwise secured by projectile housing **602**. Second stopper **618** may be secured by projectile housing **602** such that primer assembly **608** may translate within projectile housing **602** and collide with activator pin **616** such that primer charge **614** is activated. Primer charge **614** may be associated with an activation threshold that is associated with an amount of energy for activating primer charge **614**, the amount of energy being provided by primer activator **612** contacting activator pin **616**, primer activator **612** compressing primer charge **614** against activator pin **616**, and/or otherwise applying kinetic energy from primer assembly **608** to primer charge **614**. Second stopper **618** may be configured to remain secured by projectile housing **602** and enable primer charge **614** to be activated by collision of primer assembly **608** and activator pin **616**. It should be noted that second stopper **618** may be configured to remain substantially static under initial contact of primer assembly **608** and activator pin **616**.

[0138] In various embodiments, second stopper **618** may secure activator pin **616**. For example, activator pin **616** may be coupled with, attached to, and/or otherwise secured within a recess associated with second stopper **618**. Similar to the discussion above, activator pin **616** may be secured by second stopper to contact primer activator **612** during translation of primer assembly **608**. Activator pin **616** may be configured to remain secured by second stopper **618** and collide with primer activator **612**. It should be noted that activator pin **616** may be configured to remain substantially static under initial contact of primer activator **612**. Alternatively, projectile housing **602** may secure activator pin **616** during translation of primer assembly **608** to collide with primer activator **612**. Activator pin **616** and primer activator **612** may be disposed co-axially with projectile housing **602** or parallel to a central axis of projectile housing **602**.

[0139] In various embodiments, second stopper **618** may be removably secured within projectile housing **602**. As noted above, second stopper **618** may be secured such that primer charge **614** is activated in response to primer activator **612** contacting activator pin **616**. Additionally, second stopper **618** may be secured within projectile housing **602** such that the activation of primer charge **614** dislodges second stopper **618**. For example, activation of primer charge **614** may cause primer charge **614** to release an amount of expanding gases. The expanding gases may exert pressure on second stopper **618** sufficient to dislodge second stopper **618** from a first stopper position secured by projectile housing **602**. Further, the expanding gases may exert pressure on projectile payload **620** in contact via pressure applied to second stopper **618**.

[0140] In various embodiments, activator pin **616** may be removably secured by at least one of projectile housing **602** and/or second stopper **618**. As noted above, activator pin **616** may be secured such that primer activator **612** may contact activator pin **616** to activate primer charge **614**. Additionally, activator pin **616** may be dislodged after activation of primer charge **614** by pressure exerted by expanding gases released by primer charge **614**. The expanding gases may exert pressure on activator pin **616** sufficient to dislodge activator pin **616** from a first pin position.

[0141] In various embodiments, expanding gases (or another pressure source) emitted by primer charge **614** may exert pressure upon at least one of activator pin **616** and/or second stopper **618** sufficient to deploy projectile payload **620** from projectile housing **602**. As noted above, pressure exerted by primer charge **614** may dislodge, decouple, detach, and/or otherwise move activator pin **616** and/or second stopper **618** within projectile housing **602**. Translation of activator pin **616** and/or second stopper **618** may cause projectile payload to rupture, open, and/or otherwise deploy from withing projectile housing **602**. Alternatively, or in addition, activator pin **616** and/or second stopper **618** may be displaced within projectile housing **602** such that expanding gases emitted by primer charge **614** apply pressure directly to projectile payload **620**.

[0142] In various embodiments, and independent of whether pressure and/or force is applied directly or indirectly to projectile payload **620**, primer charge **614** may cause housing fault **624** to rupture, break, shatter, dislodge, and/or otherwise open to deploy projectile payload **620**. In particular, projectile payload **620** may be a fluid or a solid with fluid-like properties (e.g., a powdered solid) such that pressure applied by primer charge **614** exerts pressure on at least projectile housing **602**, impact absorber **622**, and/or housing fault **624**. Housing fault **624** may be a portion of projectile housing **602** that is configured to have a failure threshold associated with an applied pressure and/or force less than that of projectile housing **602** and/or impact absorber **622**. Primer charge **614** may apply a pressure, optionally via projectile payload **620**, sufficient to exceed the failure threshold of housing fault **624**. In response, housing fault **624** may open and permit projectile payload **620** to be deployed from projectile **600** by primer charge **614**.

[0143] In various embodiments, and with reference to FIGS. **7**A-**7**C, projectile **600** may be configured to deploy projectile payload **620** from within projectile housing **602** in response to impact absorber **622** contacting target **702**. Primer assembly **608** may be configured to deploy projectile payload **620** by dislodging activator pin **616** from a stopper channel **704**, dislodging second stopper **618**, opening projectile housing **602**, and/or otherwise applying a deploying force to projectile payload **620**.

[0144] In various embodiments, and with reference to FIG. **7**A, primer assembly **608** may be configured to deploy, disperse, and/or otherwise release projectile payload **620** from within projectile housing **602** into dispersed payload **706**. As noted above, primer assembly **608** may be configured to activate in response to impact absorber **622** contacting target **702**. Activation of primer assembly **608** may cause a deployment force to be applied to at least one of projectile housing **602**, activation pin **616**, second stopper **618**, and/or projectile payload **620**. The deployment force may dislodge activator pin **616** from second stopper **618** and cause projectile payload **620** to be dispersed into dispersed payload **706**. Further, the deployment force, when applied to at least the projectile payload **620**, may cause a portion of projectile housing **602** (e.g., housing fault **624**) to open. The deployment force may create one or more housing openings **708** to enable deployment of dispersed payload **706**.

[0145] In various embodiments, and as noted above, one or more housing openings **708** may be created by the deployment force opening one or more portions of projectile housing **602**. For example, the one or more portions of projectile housing **602** may be weakened such that the deployment force may open one or more housing openings **708** for deployment of dispersed payload **706** proximate to target **702**. As projectile **600** (and other projectiles discussed herein) are formed to deliver projectile payloads for an effect upon target **702**, projectile housing **602** and primer assembly **608** may be formed to mitigate ineffective deployment of dispersed payload **706** (e.g., deployment of dispersed payload **706** away from target **702**). Alternatively, or in addition, projectile **600** and components of projectile **600** may be formed to cause dispersed payload **706** to contact at least a portion of the target **702**. For example, dispersed payload **706** may be deployed from the one or more housing openings **708** at least radially outward from projectile **600**.

[0146] In various embodiments, one or more housing faults **624** may be located proximate to impact absorber **622** and/or a projectile tip. Alternatively, or in addition, one or more housing faults **624** may be disposed at one or more locations on projectile housing **602** (e.g., housing faults may be formed proximate to the projectile tip, the projectile base, and/or elsewhere on the projectile housing **602**). Accordingly, the deployment force may create one or more housing openings **708** at a set of one or more housing faults **624**. For example, one or more housing faults **624** may be disposed proximate to impact absorber **622**. In response to impacting target **702**, primer assembly

608 may generate the deployment force and open one or more housing faults **624** may open to form one or more housing openings **708**. Additionally, the deployment force may disperse projectile payload **620** to form dispersed payload **706** proximate to and/or contacting target **702**.

[0147] In various embodiments, and with reference to FIG. 7B, primer assembly **608** may be configured to activate and deploy, disperse, and/or otherwise release projectile payload **620** from within projectile housing **602** into dispersed payload **706**. Activation of primer assembly **608** may cause a deployment force to be applied to at least one of projectile housing **602**, activation pin **616**, second stopper **618**, projectile payload **620**, and/or impact absorber **622**. The deployment force may dislodge activator pin **616** from second stopper **618**. Alternatively, the deployment force may dislodge activator pin **616** and second stopper **618** from projectile housing **602**. Additionally, and based at least in part on activator pin **616** and/or second stopper **618** being dislodged, the deployment force may cause projectile payload **620** to be deployed to form dispersed payload **706**. Further, and based at least in part on activator pin **616** and/or second stopper **618** being dislodged, the deployment force may cause projectile housing **602** to separate from impact absorber **622**. The deployment force may create opened end **710** to enable deployment of dispersed payload **706**.

[0148] In various embodiments, the deployment force may cause impact absorber **622** to be separated from projectile housing **602** to enable deployment of projectile payload **620**. For example, and at a first time (e.g., during flight of projectile **600**), projectile housing **602** and impact absorber **622** may be coupled at external absorber surface **712** and housing internal surface **714**. Additionally, and at a second time, the deployment force may be applied to at least impact absorber **622** such that external absorber surface **712** is decoupled from housing internal surface **714**. Decoupling external absorber surface **712** from housing internal surface **714** may enable the deployment force to separate impact absorber **622** from projectile housing **602** and projectile payload **620** to be deployed from opened end **710**.

[0149] External absorber surface **712** and housing internal surface **714** may be removably coupled prior to launch from projectile launcher **100**, during launch from projectile launcher **100**, and during a flight of projectile **600** between projectile launcher **100** and target **702**. External absorber surface **712** and housing internal surface **714** may be coupled via mechanical couplings (e.g., threading, ridges, latches, etc.), adhesives, and/or other coupling mechanisms. Additionally, external absorber surface **712** and housing internal surface **714** may be decoupled by an applied force satisfying a force threshold. For example, impact absorber **622** may be dislodged from projectile housing **602** in response to the deployment force, applied by primer assembly **608**, exceeds the force threshold and decouples external absorber surface **712** from housing internal surface **714**.

[0150] In various embodiments, decoupling external absorber surface **712** from housing internal surface **714** may remove impact absorber **622** from projectile housing **602**. Removal of impact absorber **622** from projectile housing **602** may enable projectile payload **620** to be deployed from within projectile housing **602** via open end **710**. It should be noted that while open end **710** refers to the projectile tip after impact absorber **622** has been removed, an open end may refer to first end **404** after removal of first stopper **412**, second end **406** after removal of second stopper **416**, a projectile base after one or more stoppers have been removed, and/or a projectile tip after one or more stoppers have been removed. Additionally, removal of impact absorber **622** (and/or one or more stoppers disposed within projectile housing **602**) may reference external absorber surface **712** translating within internal housing surface **714** such that at least a portion of impact absorber **622** is no longer radially within projectile housing **602**.

[0151] In various embodiments, projectile payload **620** may be deployed from within projectile housing **602** via open end **710**. For example, and after impact absorber **622** has been separated and/or removed from projectile housing **602**, the deployment force may cause projectile payload **620** to translate within projectile housing **602** and through open end **710**. Translation of projectile payload **620** through open end **710** may form dispersed payload **706**.

[0152] In various embodiments, primer assembly **608** may apply the deployment force to first stopper **604**, activator pin **616**, second stopper **618**, and/or other components of projectile **600**. The deployment force may be configured to cause at least impact absorber and projectile payload **620** to be deployed from within projectile housing **602**. The deployment force may be configured to cause projectile housing **602** to translate away from target **702** such that impact absorber **622** is separated from projectile housing **602**. Additionally, the deployment force may be applied to projectile payload **620** and impact absorber **622** via second stopper **618**. Where projectile payload **620** and impact absorber **622** are substantially prevented from movement by target **702**, the deployment force may be applied to projectile housing **602** via first stopper **604**. Application of the deployment force to first stopper may initiate translation of projectile housing **602** away from target **702**. Translation of projectile housing **602** away from target **702** may cause impact absorber **622** to be separated from projectile housing **602** and deploy projectile payload **620** into dispersed payload **706**.

[0153] In various embodiments, and with reference to FIG. **7**C, primer assembly **608** may be configured to activate and rupture projectile housing **602** to deploy projectile payload **620** into dispersed payload **706**. The deployment force may cause projectile housing **602** to rupture and/or otherwise open to release projectile payload **620**. As noted above, with reference to FIGS. **5**A-**5**C, rupturing projectile housing **602** may cause projectile housing **602** to split, break, and/or otherwise separate into one or more housing portions **716**. Separation of projectile housing **602** into one or more housing portions **716** may further cause projectile payload to be dispersed and/or deployed to form dispersed payload **706**.

[0154] Projectile housing **602**, first stopper **604**, activator pin **616**, and/or second stopper **618** may be configured such that the deployment force causes projectile housing **602** to rupture. For example, first stopper **604** and second stopper **618** may be secured within projectile housing **602** such that activation of primer assembly **608** causes projectile housing **602** to separate into housing portions **716**. Additionally, activation of primer assembly **608** may generate the deployment force via expanding gases that are substantially contained within projectile housing **602** by first stopper **604** and second stopper **618**. Under the deployment force generated by the primer assembly **608**, projectile housing **602** may rupture to permit the expanding gases to be released and alleviate pressure buildup associated with the deployment force. Release of the expanding gases may further cause projectile payload **620** to be deployed into dispersed payload **706**.

[0155] In various embodiments, a projectile may be formed to be launched from a projectile launcher. The projectile may comprise a projectile housing, a projectile payload, a primer assembly, and a primer activator. The projectile housing may extend between a first end and a second end opposite the first end. The projectile housing may define a housing channel extending at least partially between the first end and the second end. Additionally, the projectile payload may be disposed within the housing channel of the projectile housing. Similarly, the primer assembly may be disposed within the housing channel. The primer assembly may be comprised of a primer charge, the primer charge configured to deploy the projectile payload from the projectile housing. The primer activator may be disposed within the housing channel and configured to activate the primer charge based at least in part on the first end of the projectile contacting a target.

[0156] In various embodiments, the projectile may be launched from the projectile launcher by a launch force towards the target to disperse the projectile payload proximate to the target. For example, the first end of the projectile may be an impact absorber that contacts the target and disperses an impact force on the target. Similarly, the second end of the projectile may receive a launch force from the projectile launcher, the launch force causing the impact absorber to contact the target. The primer assembly may be configured to translate from a first position to a second position based at least in part on the first end contacting the target. It should be noted that the first position of the primer assembly is associated with a primer lock securing the primer assembly within the housing channel. Additionally, the second position of the primer assembly may be associated with the primer charge being activated. Further, the primer lock may be associated with a release threshold that enables the primer assembly to translate from the first position to the second position.

[0157] In various embodiments, the primer lock may transition from a first state to a second state based at least on the primer assembly satisfying a release threshold. The first state of the primer lock may secure the primer assembly in the first position. The second state of the primer lock may permit the primer assembly to translate from the first position to the second position. The release threshold may be satisfied by a decelerating force generated by the first end contacting the target. Additionally, the primer assembly may translate from the first position associated with the first state to the second position based at least in part on the projectile housing being substantially arrested by contact between the first end and the target. For example, the primer assembly may be associated with and/or receive an amount of kinetic energy based at least in part on the launch force. Further, the amount of kinetic energy may cause the primer assembly to translate from the first position to the second position and activate the primer charge in the second position. The release threshold may be satisfied by the amount of kinetic energy in response to the first end contacting the target and cause the primer lock to transition from the first state to the second state. For example, the primer lock may be configured as a securing ring disposed radially outside the primer assembly that deforms, based at least in part on the release threshold being satisfied, to permit translation of the primer assembly radially within the securing ring.

[0158] In various embodiments, the primer assembly may comprise a primer activator, wherein the primer assembly translates from a first position to a second position such that the primer activator activates the primer charge in the second position. Translation of the primer assembly from the first position to the second position may cause the primer activator to collide with an activation pin to ignite the primer charge. It should be noted that activation of the primer charge may rupture at least a portion of the projectile housing to deploy the projectile payload. Alternatively, or in addition, activation of the primer charge dislodges a projectile stopper from the projectile housing to deploy the projectile payload. Additionally, the primer charge may be configured as an amount of pyrotechnic material that generates a deployment force that, when applied to the projectile payload, causes the projectile payload to be deployed from within the projectile housing.

[0159] In various embodiments, the projectile may be comprised of one or more portions that are arranged to form the projectile. For example, a first portion may comprise the first end of the projectile, wherein the first portion is a projectile base that receives a launch force and applies the launch force to at least the projectile housing. Additionally, a second portion may comprise the projectile payload, the second portion containing the projectile payload at a first time and permitting the projectile payload to be dispersed at a second time. Further, a third portion may comprise the primer assembly, the third portion containing primer assembly at a first position at the first time and at a second position at the second time. Similarly, a fourth portion may comprise the second end, the fourth portion comprising an impact absorber that causes the primer assembly to translate within the third portion from the first position to the second position. It should be noted that the projectile may comprise one or more additional portions that are configured to further augment the projectile. The first portion may be disposed at the first end of the projectile, the second portion may be disposed substantially adjacent to the first portion, the third portion may be disposed substantially adjacent to the second portion opposite the first portion, and the fourth portion may be disposed substantially adjacent to the third portion opposite the second portion. Alternatively, the first portion may be disposed at the first end, the third portion is disposed substantially adjacent to the first portion, the second portion is disposed substantially adjacent to the third portion opposite the first portion, and the fourth portion is disposed substantially adjacent to the second portion opposite the third portion. It should be noted that the projectile may comprise one or more of each the above portions (e.g., a second portion may be disposed between the first portion and the third portion and an additional second portion may be disposed between the third portion and the fourth portion).

[0160] In various embodiments, the projectile housing may comprise one or more faults that enable the primer assembly to disperse the projectile payload from within the projectile housing. Activation of the primer assembly may rupture the projectile housing at the one or more faults and may cause the projectile housing to break into one or more projectile housing portions. The primer charge may cause the projectile housing to split along the one or more faults and form the one or more projectile housing portions. Alternatively, or in addition, one or more faults may be disposed proximate to the second end of the projectile and enable the projectile payload to be dispersed from within the projectile housing proximate to the target via one or more openings, in the projectile housing, formed at the one or more faults.

[0161] In various embodiments, a primer portion of a projectile may be configured to deploy a payload from a projectile. The primer portion may comprise a primer charge, a primer activator, a primer wall, and a primer lock. The primer charge may be associated with a rest state and an active state, the active state causing the primer charge to rupture the projectile and deploy the payload from within the projectile. The primer activator may be configured to utilize an amount of kinetic energy to transition the primer charge from the rest state to the active state. The primer wall may be configured to translate between a first position and a second position. Additionally, the primer charge may be secured substantially adjacent to primer activator by the primer wall. Further, the first position may be associated with the rest state and the second position may be associated with the active state. The primer lock may be disposed between the primer wall and a projectile housing and selectively couple the primer wall to the projectile housing. Further, the primer lock may couple with the primer wall in the first position and decouple from the primer wall and/or the projectile housing to permit translation from the first position to the second position.

[0162] In various embodiments, the primer lock may secure the primer wall in the first position at a first time and permits the primer wall to translate to the second position at a second time. The primer lock may permit the primer wall to translate from the first position to the second position based at least in part on the projectile impacting a target. Additionally, the primer lock may be configured as a friction lock that secures the primer wall in the first position via contact with the primer wall and a housing of the projectile. Further, the primer lock may be associated with a force threshold that causes the primer lock to permit translation of the primer wall based at least on the amount of kinetic energy exceeding the force threshold.

[0163] In various embodiments, an amount of kinetic energy may be associated with at least the primer activator during a flight of the projectile. The amount of kinetic energy may be applied to the primer charge by the primer activator based at least in part on the projectile impacting a target, wherein application of the amount of kinetic energy may cause the primer charge to transition from the rest state to the active state. Based at least on the primer charge being activated, the primer charge may rupture the projectile housing by creating one or more openings in the projectile housing proximate to a target. Additionally, the primer charge, in the active state, may generate a dispersal force that causes the payload to disperse proximate to the target through the one or more openings. Alternatively, or in addition, the primer charge may cause an impact absorber associated with the projectile to be dislodged from the projectile housing and form an open end of the projectile housing, Further, the primer charge, in the active state, may generate a dispersal force that causes the payload to disperse through the open end.

[0164] In various embodiments, the primer wall may comprise a lock stop that substantially prevents translation from the first position to the second position. The lock stop may permit translation from the first position to the second position based at least in part on the lock stop receiving the amount of kinetic energy.

[0165] In various embodiments, a projectile launcher may be configured to launch a projectile and cause a projectile payload to be deployed, dispersed, and/or otherwise released proximate to the target. For example, the projectile launcher may launch the projectile towards a target, the projectile being associated with an amount of kinetic energy and a flight velocity. The projectile launcher may cause the projectile to impact the target, wherein an impact absorber of the projectile substantially arrests the flight velocity of the projectile. Additionally, the projectile launcher may cause, based at least on the amount of kinetic energy, a primer assembly of the projectile to activate. Further, the projectile launcher may cause a projectile payload to be dispersed proximate to the target, wherein activation of the primer assembly disperses the projectile payload from the projectile.

[0166] In various embodiments, the projectile launcher may cause the primer assembly to activate, wherein activation of the primer assembly may cause the primer assembly to translate between a first position and a second position within the projectile. Activation of primer assembly may cause the primer assembly to contact an activation pin and emit an amount of gas that disperses the projectile payload from within the projectile. Additionally, the amount of kinetic energy provided during launch of the projectile may cause the primer assembly to translate within the projectile based at least on the impact absorber substantially arresting the flight velocity. The amount of kinetic energy may result in a collision and/or contact between the primer assembly and an activation pin. The primer assembly and the activator pin may contact in the second position such that the primer assembly releases an amount of gas that disperses the projectile payload. It should be noted that the amount of kinetic energy provided by the projectile launcher may be determined to cause the primer assembly to translate from the first position to the second position in response to the impact absorber substantially arresting the projectile. Similarly, the amount of kinetic energy provided by the projectile launcher may be determined to activate the primer assembly in the second position.

[0167] In various embodiments, the projectile launcher may cause, based at least in part on the amount of kinetic energy provided during launch of the projectile, the primer assembly to activate and emit an amount of gas. Additionally, the projectile launcher may cause, based at least in part on the amount of gas released by the primer assembly, a projectile housing to rupture and disperse the projectile payload. Alternatively, or in addition, the primer assembly may activate and generate a dispersal force that causes the projectile payload to be dispersed from within the projectile housing and/or ruptures the projectile housing. Rupture of the projectile housing may cause the projectile housing to split into, break into, and/or otherwise form one or more projectile housing portions. Alternatively, rupture of the projectile housing may form one or more openings in projectile housing that enables the projectile payload to be dispersed proximate to the target.

[0168] In various embodiments, a deployment of a projectile from a projectile launcher may cause a projectile payload to be dispersed proximate to a target of the projectile launcher. For example, the projectile may receive a launch force from a propulsion module of a projectile launcher. The projectile may traverse and/or travel a distance towards the target at a flight velocity. Additionally, the projectile may contact the target with an impact absorber, wherein the impact absorber substantially arrests a projectile housing of the projectile. Further, the projectile impacting the target may cause, based at least in part on the impact absorber arresting the projectile housing, a primer assembly to translate from a first position to a second position. Translation from the first position to the second position may activate the primer assembly. In response to activation of the primer assembly, a projectile payload may be dispersed proximate to the target.

[0169] In various embodiments, the primer assembly may translate from the first position to a second position. Translation of the primer assembly within the projectile may cause the primer assembly in the second position to contact an activation pin. In response to the primer assembly contacting the activation pin, the primer assembly may apply a dispersal force to the projectile payload. For example, contact between the primer assembly and the activation pin may ignite a primer charge and causes the primer charge to generate the dispersal force. Alternatively, or in addition, contact between the primer assembly and the activation pin converts an amount of kinetic energy associated with the flight velocity of the projectile into an amount of activation energy for the primer charge. The dispersal force may be applied to the projectile housing such that the dispersal force creates one or more housing openings associated with the projectile housing. Additionally, the dispersal force may cause the projectile payload to be dispersed via the one or more housing openings. Further, the dispersal force may be applied to an internal stopper to decouple the internal stopper from the projectile housing. The dispersal force may be applied to the projectile payload via the internal stopper and cause the projectile payload to exit the projectile housing.

[0170] In various embodiments, the launch force may be applied to an internal stopper coupled to the projectile housing such that the propulsion module causes at least the projectile housing to accelerate from a rest state to the flight velocity. The flight velocity may be associated with an amount of kinetic energy that satisfies an activation threshold associated with the primer assembly. Additionally, the activation threshold may be associated with an amount of energy that causes the primer assembly to translate from the first position to the second position. Further, the activation threshold may be associated with an additional amount of energy that causes the projectile payload to be dispersed.

[0171] In various embodiments, the impact absorber may apply a decelerating force to arrest the projectile housing. Additionally, the impact absorber, based at least in part on application of the decelerating force, may cause the primer assembly to translate from the first position to the second position. Further, a primer lock may secure the primer assembly in the first position at a first time, the first time associated with the projectile housing having the flight velocity. The primer lock may decouple from the primer assembly to permit translation from the first position to the second position at a second time, the second time associated with the impact absorber arresting the projectile housing.

[0172] Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosures. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims and their legal equivalents, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B, and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C.

[0173] Systems, methods, and apparatus are provided herein. In the detailed description herein, references to “various embodiments,” “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element is intended to invoke 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

Claims

1. A projectile for a projectile launcher, the projectile comprising: a projectile housing disposed between a first end of the projectile and a second end of the projectile opposite the first end, the projectile housing comprising a housing channel extending at least partially between the first end and the second end; a projectile payload disposed within the housing channel; a primer assembly disposed within the housing channel and comprising a primer charge, the primer charge configured to deploy the projectile payload from within the projectile housing; and a primer activator disposed within the housing channel and configured to activate the primer charge based at least in part on the first end of the projectile contacting a target.
2. The projectile of claim 1, wherein: the first end comprises an impact absorber that contacts the target and disperses an impact force on the target; and the second end receives a launch force from the projectile launcher, the launch force causing the impact absorber to contact the target and receive the impact force.
3. The projectile of claim 1, wherein the primer assembly is configured to translate from a first position to a second position based at least in part on the first end contacting the target.
4. The projectile of claim 1, wherein: a first position of the primer assembly is associated with a primer lock securing the primer assembly within the housing channel; a second position of the primer assembly is associated with the primer charge being activated; and the primer lock is associated with a release threshold that enables the primer assembly to translate from the first position to the second position.
5. The projectile of claim 4, wherein: the release threshold is satisfied by a decelerating force generated by the first end contacting the target; and the primer assembly translates from the first position to the second position based at least in part on the projectile housing being substantially arrested by contact between the first end and the target.
6. The projectile of claim 4, wherein: the primer lock is configured as a securing ring disposed radially outside the primer assembly; and the securing ring deforms, based at least in part on the release threshold being satisfied, to permit translation of the primer assembly along a central axis of the projectile and within the securing ring.
7. The projectile of claim 1, wherein: the primer assembly comprises the primer activator; and the primer assembly translates from a first position to a second position such that the primer activator activates the primer charge in the second position.
8. The projectile of claim 7, wherein: translation of the primer assembly from the first position to the second position causes the primer activator to collide with an activation pin to ignite an amount of pyrotechnic material of the primer charge; and activation of the primer charge ruptures at least a portion of the projectile housing to deploy the projectile payload.
9. The projectile of claim 1, further comprising: a first portion that comprises the first end, wherein the first portion is a projectile base that receives a launch force and applies the launch force to at least the projectile housing; a second portion that comprises the projectile payload, the second portion containing the projectile payload at a first time and permitting the projectile payload to be dispersed at a second time; a third portion that comprises the primer assembly, the third portion containing primer assembly at a first position at the first time and at a second position at the second time; and a fourth portion that comprises the second end, the fourth portion comprising an impact

absorber that causes the primer assembly to translate within the third portion from the first position to the second position.

10. The projectile of claim 9, wherein: the first portion is disposed at the first end; the second portion is disposed substantially adjacent to the first portion; the third portion is disposed substantially adjacent to the second portion opposite the first portion; and the fourth portion is disposed substantially adjacent to the third portion opposite the second portion.

11. The projectile of claim 9, wherein: the first portion is disposed at the first end; the third portion is disposed substantially adjacent to the first portion; the second portion is disposed substantially adjacent to the third portion opposite the first portion; and the fourth portion is disposed substantially adjacent to the second portion opposite the third portion.

12. The projectile of claim 1, wherein the projectile housing comprises one or more faults that enable the primer assembly to disperse the projectile payload from within the projectile housing by rupturing the projectile housing at the one or more faults.

13. A primer portion of a projectile, the primer portion comprising: a primer charge associated with a rest state and an active state, the active state causing the primer charge to rupture the projectile and deploy a payload from within the projectile; a primer activator configured to utilize an amount of kinetic energy to transition the primer charge from the rest state to the active state; a primer wall configured to translate between a first position and a second position; and a primer lock disposed between the primer wall and a projectile housing, the primer lock configured to selectively couple the primer wall to the projectile housing in the first position and decouple the primer wall from the projectile housing to permit translation from the first position to the second position, wherein: the primer charge is secured substantially adjacent to primer activator by the primer wall; and the first position is associated with the rest state and the second position is associated with the active state.

14. The primer portion of claim 13, wherein the primer lock secures the primer wall and the primer charge in the first position at a first time and permits the primer wall to translate past the primer lock to the second position at a second time based at least on the projectile impacting a target.

15. The primer portion of claim 13, wherein: the amount of kinetic energy is associated with at least the primer activator during a projectile flight; and the amount of kinetic energy is applied to the primer charge by the primer activator based at least in part on the projectile impacting a target, application of the amount of kinetic energy causing the primer charge to transition from the rest state to the active state.

16. The primer portion of claim 13, wherein: the primer wall comprises a lock stop that, at a first time, substantially prevents translation past the primer lock from the first position to the second position; and the lock stop, at a second time after the first time and based at least in part on receiving the amount of kinetic energy, causes the primer lock to permit translation of the primer wall from the first position to the second position.

17. The primer portion of claim 13, wherein: the primer charge, based at least in part on entering the active state, ruptures the projectile housing; and the primer charge, based at least in part on the projectile housing being ruptured, disperses the payload from within the projectile housing.

18. A method performed by a projectile, the method comprising: receiving a launch force from a propulsion module of a projectile launcher; traversing, at a flight velocity, towards a target of the projectile launcher; contacting the target with an impact absorber of the projectile, wherein the impact absorber substantially arrests a projectile housing of the projectile; causing, based at least in part on the impact absorber arresting the projectile housing, a primer assembly of the projectile to translate from a first position to a second position, wherein translation from the first position to the second position activates the primer assembly; and causing, based at least in part on activation of the primer assembly, a projectile payload of the projectile to be dispersed proximate to the target.

19. The method of claim 18, wherein causing the primer assembly to translate from the first position to a second position further comprises: causing the primer assembly in the second position to contact an activation pin of the projectile; and causing, based at least in part on the primer assembly contacting the activation pin, the primer assembly to apply a dispersal force to the projectile payload.

20. The method of claim 19, wherein the dispersal force is applied to at least one of: the projectile housing to deploy the projectile payload by rupturing the projectile housing; one or more faults of the projectile housing to create one or more openings for deploying the projectile payload; or an internal stopper of the projectile to cause the internal stopper to be decoupled from the projectile housing and deploy the projectile payload.